Lagrangians

ex: $\mathcal{L} = \frac{1}{2} m \left(v_x^2 + v_y^2 \right)$

switch to (r, θ) as coords

$x = r \cos\theta$ $\qquad$ $\dot{x} = \dot{r} \cos\theta - r \sin\theta \, \dot{\theta}$

$y = r \sin\theta$ $\qquad$ $\dot{y} = \dot{r} \sin\theta + r \cos\theta \, \dot{\theta}$

$\dot{x}^2 + \dot{y}^2 = \dot{r}^2 + r^2 \dot{\theta}^2$ $\qquad$ $\mathcal{L} = \frac{1}{2} m \left(v_r^2 + r^2 v_\theta^2 \right)$

EL

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial v_x} - \frac{\partial \mathcal{L}}{\partial x} = 0 \quad \rightarrow \quad \ddot{x} = 0$$

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial v_y} - \frac{\partial \mathcal{L}}{\partial y} = 0 \quad \rightarrow \quad \ddot{y} = 0$$

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial v_r} \left(r(t), \theta(t), \dot{r}(t), \dot{\theta}(t) \right) - \frac{\partial \mathcal{L}}{\partial r} \left(r(t), \theta(t), \dot{r}(t), \dot{\theta}(t) \right) \rightarrow$$

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial v_\theta} \left(r(t), \theta(t), \dot{r}(t), \dot{\theta}(t) \right) - \frac{\partial \mathcal{L}}{\partial \theta} \left(r(t), \theta(t), \dot{r}(t), \dot{\theta}(t) \right) \rightarrow$$

$$\frac{\partial \mathcal{L}}{\partial v_r} = m v_r \qquad \frac{\partial \mathcal{L}}{\partial v_\theta} = m r^2 v_\theta \qquad \frac{\partial \mathcal{L}}{\partial r} = m r v_\theta^2$$

$$\frac{d}{dt} (m \dot{r}) - m r \dot{\theta}^2 = 0 \qquad \Bigg\| \qquad \frac{d}{dt} \left(m r^2 \dot{\theta} \right) = 0$$

→ Conservation of angular momentum

minimization problem

- in calculus if $f : U \subseteq \mathbb{R} \to \mathbb{R}$, the minima/maxima of f satisfy $f'(x) = 0$

generalize to functionals (functions of functions)

$L(\gamma)$ length of curve $\gamma : [0,1] \to \mathbb{R}^2$

Lagrangian

in Euclidean space $L(\gamma) = \int_0^1 \| \dot{\gamma}(t) \| \, dt$

action

Solve using variations

back to calculus

a variation of the point $x \in \mathbb{R}$ is a function $\varepsilon \in (-1,1) \mapsto x_\varepsilon \in \mathbb{R}$

s.t. $x_0 = x$

$f'(x) = 0 \iff \frac{d}{d\varepsilon} f(x_\varepsilon)\Big|_{\varepsilon=0} = 0$ for all variations x_ε

$f'(x_0) \frac{dx_\varepsilon}{d\varepsilon}\Big|_{\varepsilon=0} = 0$ for all variations

$\Rightarrow f'(x_0) = 0$

---

generalize if S is a functional on some space of functions $\{\gamma : [t_0, t_1] \to M\}$

variation of γ: $(\varepsilon, t) \in (-1,1) \times [t_0, t_1] \mapsto \gamma_\varepsilon(t) \in M$ (homotopy)

can also think of γ_ε as a curve for each ε

require $\gamma_0 = \gamma$.

γ is a critical point of S if $\frac{d}{d\varepsilon}\Big|_{\varepsilon=0} S(\gamma_\varepsilon) = 0$ for all variations of γ

ex $\gamma : [0,1] \to \mathbb{R}$ $\qquad S(\gamma) = \int_0^1 \gamma(t)\, dt$

γ_ε arbitrary variation $\qquad \frac{d}{d\varepsilon} S(\gamma_\varepsilon)\Big|_{\varepsilon=0} = \int_0^1 \frac{d}{d\varepsilon}\gamma_\varepsilon(t)\Big|_{\varepsilon=0} dt$ not always 0

$\to$ arbitrary

no critical points

$S(\gamma) = \int_0^1 \gamma(t)^2\, dt$

$\frac{d}{d\varepsilon} S(\gamma_\varepsilon)\Big|_{\varepsilon=0} = \int_0^1 \frac{d}{d\varepsilon}\gamma_\varepsilon^2(t)\Big|_{\varepsilon=0} dt = \int_0^1 2\gamma(t) \frac{d}{d\varepsilon}\gamma_\varepsilon(t)\Big|_{\varepsilon=0} dt = 0$ for all variations

if $\gamma(t) = 0 \ \forall t$

---

fix endpoints: $\qquad \gamma : [t_0, t_1] \to \mathbb{R}$ $\qquad$ fix $\gamma(t_0) = a \qquad \gamma(t_1) = b$

variation with fixed endpoints: $\gamma_\varepsilon(t_0) = \gamma(t_0) \qquad \gamma_\varepsilon(t_1) = \gamma(t_1) \ \forall \varepsilon$

A special subclass of variational problems look like this:

$$S(\gamma) = \int_{t_0}^{t_1} \mathcal{L}(t, \gamma(t), \dot\gamma(t), \ddot\gamma(t), \dots)\, dt \qquad \gamma : [t_0, t_1] \to \mathbb{R}^n$$

$$\mathcal{L} : \mathbb{R} \times \mathbb{R}^n \times \mathbb{R}^n \times \dots \times \mathbb{R}^n \to \mathbb{R} \qquad \text{(Lagrangian)}$$

in physics generally we stop at 1 derivative

<u>Hamilton's principle:</u> the physical path is a critical point of S.

<u>Euler Lagrange eqs:</u> $\left.\dfrac{d}{d\varepsilon} S(\gamma_\varepsilon)\right|_{\varepsilon=0} = 0 \quad \forall$ variations

$$\mathcal{L} = \mathcal{L}(t, q, v) \qquad q \in \mathbb{R}^n$$

$$S(\gamma_\varepsilon) = \int_{t_0}^{t_1} \mathcal{L}(t, \gamma_\varepsilon(t), \dot\gamma_\varepsilon(t))\, dt$$

$$\left.\frac{d}{d\varepsilon} S(\gamma_\varepsilon)\right|_{\varepsilon=0} = \int_{t_0}^{t_1} \left.\frac{d}{d\varepsilon} \mathcal{L}(t, \gamma_\varepsilon(t), \dot\gamma_\varepsilon(t))\right|_{\varepsilon=0} dt$$

$$= \int_{t_0}^{t_1} \left[\frac{\partial \mathcal{L}}{\partial q^\mu}(t, \gamma(t), \dot\gamma(t)) \left.\frac{d}{d\varepsilon} \gamma^\mu_\varepsilon(t)\right|_{\varepsilon=0} + \frac{\partial \mathcal{L}}{\partial v^\mu}(t, \gamma(t), \dot\gamma(t)) \left.\frac{d\, \dot\gamma^\mu_\varepsilon(t)}{d\varepsilon}\right|_{\varepsilon=0} \right] dt$$

$$'' \; \frac{d}{dt}\left(\left.\frac{d}{d\varepsilon} \gamma^\mu(t) \right)\right|_{\varepsilon=0}$$

$$\frac{d}{d\varepsilon} \gamma^\mu_\varepsilon(t) = \delta q^\mu(t)$$

integration by parts

$$= \int_{t_0}^{t_1} \left[\frac{\partial \mathcal{L}}{\partial q^\mu}(t, \gamma(t), \dot\gamma(t))\, \delta q^\mu(t) - \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial v^\mu}(t, \gamma(t), \dot\gamma(t)) \right) \delta q^\mu(t) \right] + \left[\frac{\partial \mathcal{L}}{\partial v^\mu}(t, \gamma(t), \dot\gamma(t))\, \delta q^\mu(t) \right]_{t_0}^{t_1}$$

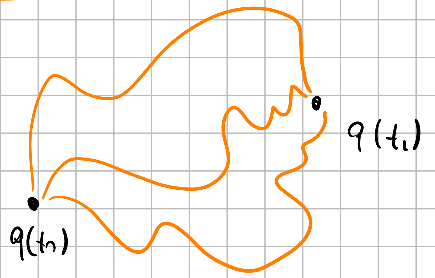
① γ is a critical point, then $\left.\dfrac{d}{d\varepsilon} S(\gamma_\varepsilon)\right|_{\varepsilon=0} = 0$ for all variations

$\to$ so it's zero for all variations with fixed end points

$$\text{so } \left[\frac{\partial \mathcal{L}}{\partial v^\mu}(t, \gamma(t), \dot\gamma(t))\, \delta q^\mu(t) \right]_{t_0}^{t_1} = 0$$

so $\int_{t_0}^{t_1} \left[\frac{\partial \mathcal{L}}{\partial q^\mu}(t,\gamma(t),\dot\gamma(t))\, \delta q^\mu(t) - \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial s^\mu}(t,\gamma(t),\dot\gamma(t)) \right) \delta q^\mu(t) \right] dt = 0 \quad \forall\, \delta q^\mu$

$$\Leftrightarrow \boxed{\frac{\partial \mathcal{L}}{\partial q^\mu}(t,\gamma(t),\dot\gamma(t)) - \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial s^\mu}(t,\gamma(t),\dot\gamma(t)) \right) = 0}$$



general form: $\mathcal{L} = \mathcal{L}(t, q_0, q_1, q_2, q_3, \dots, q_n)$

$$S = \int_{t_0}^{t_1} \mathcal{L}(t, \gamma(t), \dot\gamma(t), \ddot\gamma(t), \dots)\, dt$$

EL: $$\boxed{\sum_{k=0}^{n} (-1)^k \frac{d^k}{dt^k} \frac{\partial \mathcal{L}}{\partial q^\mu_k}(t, \gamma(t), \dot\gamma(t), \dots) = 0}$$

n=2 $$\frac{\partial \mathcal{L}}{\partial q^0} - \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial q^1} + \frac{d^2}{dt^2}\frac{\partial \mathcal{L}}{\partial q^2} = 0$$

---

adding a total derivative to the Lagrangian does not change the result

often physicists write $\mathcal{L} = \frac{1}{2} m \dot x^2 - \frac{1}{2} k x^2$

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot x} - \frac{\partial \mathcal{L}}{\partial x} = 0$$

proper way: $\mathcal{L} = \frac{1}{2} m s^2 - \frac{1}{2} k x^2$ $\qquad \mathcal{L}: \mathbb{R}^2 \to \mathbb{R}$, $(x,s) \mapsto \mathcal{L}(x,s)$

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial s}(x(t), \dot x(t)) \right) - \frac{\partial \mathcal{L}}{\partial x}(x(t), \dot x(t))$$

$$\frac{d}{dt}(m \dot x) + k x = 0 \;\rightarrow\; m \ddot x = -k x$$

if $\mathcal{L} \to \mathcal{L} + \frac{d}{dt}(f(x,\dot{x}))$ then same equations (what physicists say)

what they mean: if you add $F(x,v,a)$ to $\mathcal{L}$

s.t. $F(x(t), \dot{x}(t), \ddot{x}) = \frac{d}{dt}\left(f(x(t), \dot{x}(t))\right)$ then EL eq. are the same

$$\tilde{\mathcal{L}}(t,q,v,a) = \mathcal{L}(t,q,v) + \frac{\partial f}{\partial q} v + \frac{\partial f}{\partial v} a \qquad \text{where } f = f(q,v)$$

<u>EL:</u> $$\frac{\partial \tilde{\mathcal{L}}}{\partial q}(t,x,\dot{x},\ddot{x}) - \frac{d}{dt}\left(\frac{\partial \tilde{\mathcal{L}}}{\partial v}(t,x,\dot{x},\ddot{x})\right) + \frac{d^2}{dt^2}\left(\frac{\partial \tilde{\mathcal{L}}}{\partial a}(t,x,\dot{x},\ddot{x})\right) = 0$$

$$\frac{\partial \mathcal{L}}{\partial q} + \cancel{\frac{\partial^2 f}{\partial q^2}\dot{x}} + \cancel{\frac{\partial^2 f}{\partial q \partial v}\ddot{x}} - \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial v} + \cancel{\frac{\partial^2 f}{\partial q \partial v} v} + \cancel{\frac{\partial f}{\partial q}} + \cancel{\frac{\partial^2 f}{\partial v^2} a}\right)$$
$$+ \frac{d^2}{dt^2}\left(\cancel{\frac{\partial f}{\partial v}}\right)$$

---

geometrically $\mathcal{L} = \mathcal{L}(q,v)$ $\quad q \in M$ manifold

$v \in T_q M$

$$\mathcal{L} : TM \to \mathbb{R}$$

$\ddot{x} = 0 \iff \dot{v} = 0$, $\dot{x} = v$

$M = \mathbb{R}^2$ $\qquad \mathcal{L}(q,v) = \frac{1}{2} m \|v\|^2$

$v = v_x \frac{\partial}{\partial x} + v_y \frac{\partial}{\partial y}$ $\qquad \|v\|^2 = v_x^2 + v_y^2$ $\qquad \left\langle \frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j} \right\rangle = \delta_{ij}$

<u>change coordinates:</u> r, θ $\qquad x = r\cos\theta$

$y = r\sin\theta$

$dx = dr\cos\theta - r\sin\theta\, d\theta$

$dy = dr\sin\theta + r\cos\theta\, d\theta$

$$\frac{\partial}{\partial r} = \frac{\partial x}{\partial r}\frac{\partial}{\partial x} + \frac{\partial y}{\partial r}\frac{\partial}{\partial y}$$

$$\frac{\partial}{\partial \theta} = \frac{\partial x}{\partial \theta}\frac{\partial}{\partial x} + \frac{\partial y}{\partial \theta}\frac{\partial}{\partial y}$$

$\mathbb{R}^4 = \mathbb{R} \times \mathbb{R}^3$

path integral approach

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - m^2\phi^2$$

$$S[\phi] = \int_{\mathbb{R}^4} d^4x\, \mathcal{L}(\phi, \partial\phi)$$

$\phi: \mathbb{R}^4 \to \mathbb{R}$

$$Z = \int D\phi\; e^{iS[\phi]}$$

translation invariant

2 problems: ① there is no Lebesgue measure on set of all fields

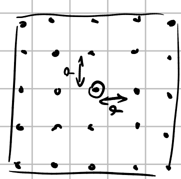
② exponential oscillates

---

to solve problem ① we can discretize spacetime

we lattice $\Lambda = \left\{ \sum_\mu a\lambda^\mu e_\mu,\ \lambda^\mu = -N, -N+1, \dots, N-1, N \right\}$ $\quad a = \frac{L}{N}$

$= \{x_i\}_{i\in I}$



$$\hookrightarrow Z = \prod_i \int_{\mathbb{R}} d\phi_i\; e^{iS[\phi]}$$

② "solve" it by doing a Wick rotation: $t \to it$

$$i\int_{\mathbb{R}^4} d^4x\, \frac{1}{2}\left((\partial_t\phi)^2 - (\nabla\phi)^2 \right)$$

$dt \to i\,dt$

$\frac{\partial}{\partial t} \to -i\frac{\partial}{\partial t}$

$$\to i^2\int_{\mathbb{R}^4} d^4x\, \frac{1}{2}\left(-(\partial_t\phi)^2 - (\nabla\phi)^2 \right)$$

$$= \frac{1}{2}\int_{\mathbb{R}^4} d^4x\, \left((\partial_t\phi)^2 + (\nabla\phi)^2 \right)$$

integration by parts

$$= -\frac{1}{2}\int_{\mathbb{R}^4} d^4x\, \left(\phi\,\partial_t^2\phi + \phi\,\nabla^2\phi \right)$$

$\nabla\phi\cdot\nabla\phi = \nabla\cdot(\phi\nabla\phi) - \phi\nabla^2\phi$

$$e^{-\frac{1}{2}\int_{\mathbb{R}^4} d^4x\,(\phi L \phi)}$$

$L = \partial_t^2 + \nabla^2$ linear differential operator

→ probability measure

multi-dimensional Gaussian on $\mathbb{R}^n$

$$e^{-\frac{1}{2}x^t A x}$$

A matrix (symmetric + pos. definite)

<u>Gaussian distribution:</u> (1D) $\quad p(x) = e^{-\frac{1}{2}x^2}$

$$\langle f \rangle = \frac{\int_{-\infty}^{\infty} dx\, e^{-\frac{1}{2}x^2} f(x)}{\int_{-\infty}^{\infty} dx\, e^{-\frac{1}{2}x^2}}$$

nD version $\quad p(\vec{x}) = e^{-\frac{1}{2}x^t A x}$

$$\langle x^i \rangle = \frac{\int_{\mathbb{R}^n} d^n x\, e^{-\frac{1}{2}x^t A x}\, x^i}{\int_{\mathbb{R}^n} d^n x\, e^{-\frac{1}{2}x^t A x}}$$

$$\langle \underbrace{\phi(x)}_{\in\mathbb{R}^4} \rangle = \frac{\int D\phi\, e^{S[\phi]}\,\phi(x)}{\int D\phi\, e^{S[\phi]}} = 0 \qquad \text{(1-point function)}$$

quantization step

(interpretation of $\langle \phi(x^1)\phi(x^2)\phi(x^3)\dots\rangle$

$$\langle 0|\,\hat{\phi}(t,\vec{x})\,|0\rangle$$

<u>Theorem:</u> <u>Isserlis theorem</u>

For a gaussian distribution $\quad p(x) = e^{-\frac{1}{2}x^t A x} \qquad x \in \mathbb{R}^n$

$$\langle x^{i_1} x^{i_2} \cdots x^{i_k} \rangle = \frac{\int_{\mathbb{R}^n} d^n x\, p(x)\, x^{i_1}\cdots x^{i_k}}{\int_{\mathbb{R}^n} d^n x\, p(x)}$$

$$= \sum_{\sigma \in P_k^2} \prod_{\{r,s\} \in \sigma} \langle x^{i_r} x^{i_s} \rangle$$

P_k^2 set of all partitions into pairs of $\{1, \dots, k\}$

Corollary: $= 0$ if k is odd

ex: $k = 4$ $\{1, 2, 3, 4\}$

partitions:

$(1,2)\ (3,4)$

$(1,3)\ (2,4)$

$(1,4)\ (2,3)$

$$\langle x^{i_1} \dots x^{i_4} \rangle = \langle x^{i_1} x^{i_2} \rangle \langle x^{i_3} x^{i_4} \rangle + \langle x^{i_1} x^{i_3} \rangle \langle x^{i_2} x^{i_4} \rangle + \langle x^{i_1} x^{i_4} \rangle \langle x^{i_2} x^{i_3} \rangle$$

for path integral:

$$\langle \phi(x^{i_1}) \dots \phi(x^{i_n}) \rangle = \sum_{\sigma \in P_k^2} \prod_{\{r,s\} \in \sigma} \langle \phi(x^{i_r}) \phi(x^{i_s}) \rangle$$

figure out the 2-point function

$$\underbrace{\langle 0 | \hat{\phi}(x)}\ \underbrace{\hat{\phi}(x') | 0 \rangle}$$

for a Gaussian

$$\langle x^i x^j \rangle = \frac{\int_{\mathbb{R}^n} d^n x\, e^{-\frac{1}{2} x^t A x}\, x^i x^j}{\int_{\mathbb{R}^n} d^n x\, e^{-\frac{1}{2} x^t A x}} = (A^{-1})_{ij}$$

for path integrals $\langle \phi(x) \phi(x') \rangle := (L^{-1})(x, x')$

$$P(\phi) = e^{-\frac{1}{2} \int_{\mathbb{R}^4} d^4x\, (\phi L \phi)}$$

$L^{-1}(x, x') = G(x, x')$ s.t $L\, G(x, x') = \delta(x - x')$ Green's function

$\xrightarrow{FT}$ $(k_0^2 + \vec{k}^2)\, \hat{G}(k) = 1$

in our case $L = \partial_t^2 + \nabla^2$ $L^{-1} = \dfrac{1}{\partial_t^2 + \nabla^2}$

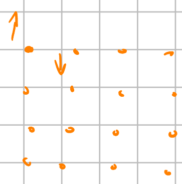
Fourier transform: $\dfrac{1}{k_0^2 + \vec{k}^2}$

in QM can do the same

$$Z[q] = \int Dq \, e^{\frac{i}{\hbar} S[q]}$$ "sum over all paths"

Statistical mechanics

$$Z = \sum_i e^{-\beta E_i}$$



Connection (path integral $\leftrightarrow$ canonical quantization)

$$\boxed{\langle \phi(\underbrace{x_1}_{\in \mathbb{R}^4}) \phi(x_2) \cdots \phi(x_n) \rangle \equiv \langle 0 | \underbrace{T}_{\text{time ordered}} (\hat{\phi}(\overset{\in \mathbb{R}^4}{x_1}) \cdots \hat{\phi}(x_n)) | 0 \rangle}$$

$$\boxed{[\hat{\phi}(t,\vec{x}), \hat{\phi}(t,\vec{y})] = 0 \qquad [\hat{q}_i, \hat{q}_j] = 0}$$

$[\hat{\phi}(t,\vec{x}), \phi(t',\vec{x}')] = 0$ if $(t,\vec{x})$, $(t',\vec{x}')$ are spacelike separated

interactions

$$S[\phi] = \underbrace{\int_{\mathbb{R}^4} d^4x \, \mathcal{L}_{free}(\phi, \partial\phi)}_{S_0[\phi]} + g \int_{\mathbb{R}^4} d^4x \, \mathcal{L}_{int}(\phi, \partial\phi)$$

$$S_0[\phi] + g \, S_I[\phi]$$

trick

$$Z[\phi] = \int D\phi \, e^{iS_0[\phi] + igS_I[\phi]}$$

$$\int d^3x \left(\alpha_3 \phi^3(x) + \alpha_4 \phi^4(x) + \dots \right)$$

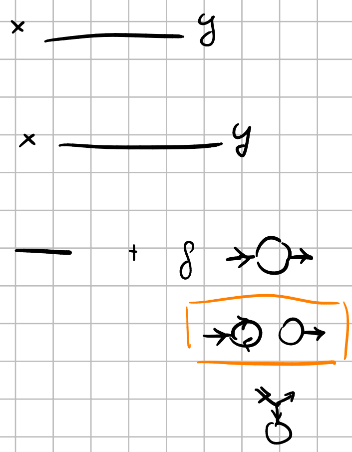
$$= \sum_{k=0}^{\infty} \frac{(ig)^k}{k!} \int D\phi \, e^{iS_0[\phi]} (S_I(\phi))^k$$

$$\langle \phi(x) \phi(y) \rangle = \frac{\sum_{k=0}^{\infty} \frac{(ig)^k}{k!} \int D\phi \, e^{iS_0[\phi]} (S_I(\phi))^k \phi(x) \phi(y)}{\sum_{k=0}^{\infty} \frac{(ig)^k}{k!} \int D\phi \, e^{iS_0[\phi]} (S_I(\phi))^k}$$

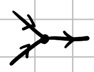
$O(1)$: $\langle \phi(x)\phi(y)\rangle_0$ ← feynman

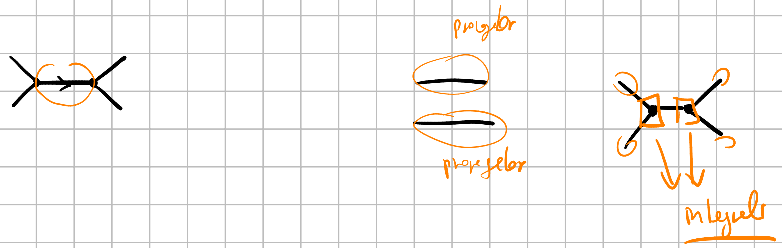
$O(g)$: $\langle \phi(x)\phi(y)\rangle_0$

$O(g^2)$:



3-point





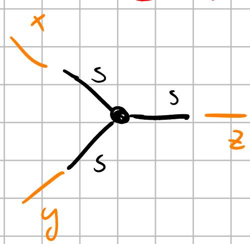
$$\langle \phi(x)\phi(y)\phi(z)\rangle = \frac{\sum_{k=0}^{\infty} \frac{(ig)^k}{k!} \int D\phi \; e^{iS_0[\phi]} \left(S_I(\phi)\right)^k \phi(x)\,\phi(y)}{\sum_{k=0}^{\infty} \frac{(ig)^k}{k!} \int D\phi \; e^{iS_0[\phi]} \left(S_I(\phi)\right)^k}$$

expand to $o(g)$

$$= \underbrace{\langle \phi(x)\phi(y)\phi(z)\rangle_0}_{=0} + \boxed{ig \int d^4(s) \; \langle \phi(x)\phi(y)\phi(z)\,\phi^3(s)\rangle_0}$$

$$- ig \underbrace{\langle \phi(x)\phi(y)\phi(z)\rangle_0}_{=0} \int d^4(s) \; \langle \phi^3(s)\rangle_0$$

Something wrong here



$1\ 2\ 3\ 4\ 5\ 6 =$ $(12)(34)(56)$

$(12)(35)(46)$

$(13)(24)(56)$

$(13)(25)(46)$

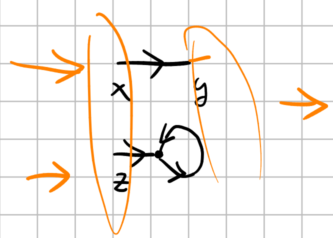
$(14)(23)$

$\langle \phi(x)\phi(y)\rangle \langle \phi(z)\,\phi(s)\rangle \langle \phi(s)^2\rangle$

$\langle \phi(x)\phi(z)\rangle \langle \phi(y)\,\phi(s)\rangle \langle \phi(s)^2\rangle$

$\langle \phi(y)\phi(z)\rangle \langle \phi(x)\,\phi(s)\rangle \langle \phi(s)^2\rangle$

$\langle \phi(x)\,\phi(s)\rangle \langle \phi(y)\,\phi(s)\rangle \langle \phi(z)\,\phi(s)\rangle$



$\langle \phi(x) \rangle =$ 

momentum space

$$\varphi(p) = \int \frac{d^4x}{(2\pi)^2}\, \phi(x)\, e^{-ip\cdot x}$$

$$\phi(x) = \int \frac{d^4p}{(2\pi)^2}\, \varphi(p)\, e^{+ip\cdot x}$$

$$S[\phi] = -\frac{1}{2}\int d^4x\, \phi \nabla^2 \phi \quad + \quad g\int d^4x\, \phi^3(x)$$

$$= \frac{1}{2}\int d^4x \int d^4p_1 \int d^4p_2\, \frac{1}{(2\pi)^4}\, \varphi(p_1)\varphi(p_2)\, p_2^2\, e^{i(p_1+p_2)\cdot x} \qquad = (2\pi)^4\,\delta(p_1+p_2)$$

$$+\, g \int d^4(x) \int d^4p_1\, d^4p_2 \int d^4p_3\, \frac{1}{(2\pi)^6}\, \varphi(p_1)\varphi(p_2)\varphi(p_3)\, e^{i(p_1+p_2+p_3)\cdot x} \qquad = (2\pi)^4\,\delta(p_1+p_2)$$

$$= \frac{1}{2}\int d^4p\, \left(-p^2\, \varphi(p)\varphi(-p)\right) + \frac{g}{(2\pi)^2}\int d^4p_1\, d^4p_2\, d^4p_3\, \varphi(p_1)\varphi(p_2)\varphi(p_3)\, \delta(p_1+p_2+p_3)$$

$$\langle \varphi(p_1) \cdots \varphi(p_k)\rangle = \underline{\sum_{\ell=0}^{\infty} g^k \int D\varphi\, e^{-S_0[\varphi]}\, S_I([\varphi])\, \varphi(p_1)\cdots\varphi(p_k)}$$

$$\frac{1}{2}\int d^4p\, \left(-p^2\, \varphi(p)\varphi(-p)\right)$$

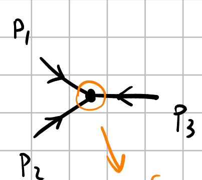
$$\downarrow$$

$$\frac{1}{2}\int d^4p_1\, d^4p_2\, \left(-p_1\cdot p_2\, \varphi(p_1)\varphi(p_2)\right)\, \delta(p_1+p_2)$$

$$\langle \varphi(p_1)\varphi(p_2)\rangle_0 = \frac{1}{(p_1+p_2)^2}$$

s_{12}

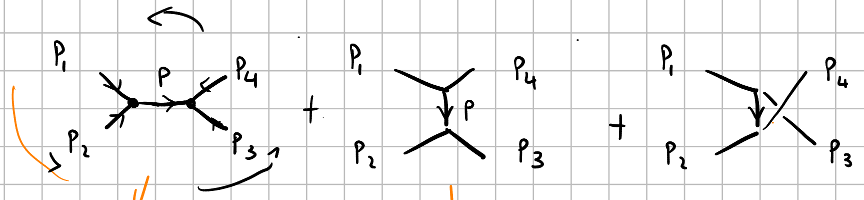
$$s_{ij} = (p_i+p_j)^2$$

$\langle \varphi(p_1)\varphi(p_2)\varphi(p_3)\rangle =$ 

$+\ O(g^2)$

$\delta(p_1+p_2+p_3)\, g$

$$\langle \varphi(p_1)\varphi(p_2)\varphi(p_3)\varphi(p_4)\rangle =$$



$$\boxed{\frac{1}{s_{12}} \qquad \frac{1}{s_{14}}} \qquad \frac{1}{s_{13}}$$

sum over permutations of only non-cherry ones

$$2\left(\frac{1}{s_{23}}+\frac{1}{s_{34}}\right)+\left(\frac{1}{s_{12}}+\frac{1}{s_{14}}\right)+\left(\frac{1}{s_{12}}+\frac{1}{s_{13}}\right)+\left(\frac{1}{s_{24}}+\frac{1}{s_{14}}\right)+\left(\frac{1}{s_{34}}+\frac{1}{s_{14}}\right)$$

$$2\left(\frac{1}{12}\ \frac{1}{14}\right)+\left(\frac{1}{12}\ \frac{1}{13}\right)+\left(\frac{1}{14}\ \frac{1}{13}\right)$$

$$8\frac{1}{s_{12}}+8\frac{1}{s_{14}}+4\frac{1}{s_{13}}$$

---

eq 2

$$\mathcal{A}_n = \underbrace{\sum_{\sigma\in S_n/Z_n}}\ \overset{\text{ordered}}{A_n}(\sigma(1),\dots,\sigma(n)) \longrightarrow \quad 1\,2\,3\,4 \sim 2\,3\,4\,1$$

$$A_4(1,2,3,4) = \frac{1}{s_{12}}+\frac{1}{s_{14}}$$

12	14
12	13
13	12
13	14
14	13
14	12

$$4\left(\frac{1}{s_{12}}+\frac{1}{s_{13}}+\frac{1}{s_{14}}\right)$$

✓ 1 2 3 4
1 2 4 3
1 3 4 2
1 3 2 4
1 4 2 3
1 4 3 2

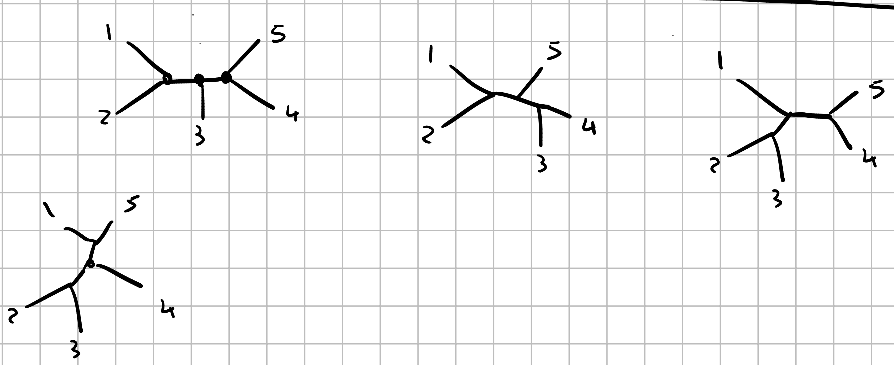
~~2 1 3 4~~
~~2 1 4 3~~
~~2 3 1 4~~
~~2 3 4 1~~
~~2 4 1 3~~
~~2 4 3 1~~

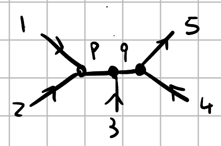
~~3 1 2 4~~
~~3 1 4 2~~
~~3 2 1 4~~
~~3 2 4 1~~
~~3 4 1 2~~
~~3 4 2 1~~

~~4 1 2 3~~
~~4 1 3 2~~
~~4 2 1 3~~
~~4 2 3 1~~
~~4 3 1 2~~
4 3 2 1

---

5 point



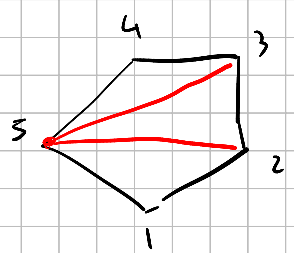


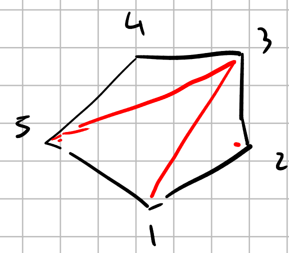
$$\int d^3q \int d^3p \, \frac{1}{p^2} \, \delta(p_1+p_2-p) \, \delta(p-q+p_3) \, \frac{1}{q^2} \, \delta(p+p_3-q) \, \delta(q+p_5+p_4)$$

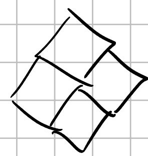
$$\frac{1}{(p_1+p_2)^2} \, \frac{1}{(p_4+p_5)^2} = \frac{\delta(p_1+p_2+p_3+p_4)}{s_{12}\, s_{45}}$$

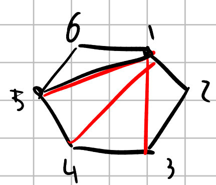
$$\frac{1}{s_{12}\, s_{45}} + \frac{1}{s_{12}\, s_{34}} + \frac{1}{s_{23}\, s_{45}} + \frac{1}{s_{23}\, s_{15}} + \frac{1}{\boxed{s_{34}\, s_{15}}}$$

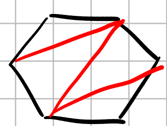
$$s_{ij} = (p_i + p_j)^2 = 2 p_i \cdot p_j$$

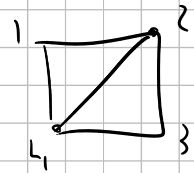












$$\frac{\partial}{\partial r} = \cos\theta \frac{\partial}{\partial x} + \sin\theta \frac{\partial}{\partial y}$$

$$v = v_r \frac{\partial}{\partial r} + v_\theta \frac{\partial}{\partial \theta}$$

$$\frac{\partial}{\partial \theta} = -r\sin\theta \frac{\partial}{\partial x} + r\cos\theta \frac{\partial}{\partial y}$$

$$\|v\|^2 = ?$$

$$\left\langle \frac{\partial}{\partial r}, \frac{\partial}{\partial r} \right\rangle = \left\langle \cos\theta \frac{\partial}{\partial x} + \sin\theta \frac{\partial}{\partial y}, \cos\theta \frac{\partial}{\partial x} + \sin\theta \frac{\partial}{\partial y} \right\rangle$$

$$= (\cos\theta)^2 + (\sin\theta)^2 = 1$$

$$\left\langle \frac{\partial}{\partial r}, \frac{\partial}{\partial \theta} \right\rangle = \left\langle \cos\theta \frac{\partial}{\partial x} + \sin\theta \frac{\partial}{\partial y}, -r\sin\theta \frac{\partial}{\partial x} + r\cos\theta \frac{\partial}{\partial y} \right\rangle$$

$$= -r\cos\theta\sin\theta + r\cos\theta\sin\theta = 0$$

$$\left\langle \frac{\partial}{\partial \theta}, \frac{\partial}{\partial \theta} \right\rangle = \left\langle -r\sin\theta \frac{\partial}{\partial x} + r\cos\theta \frac{\partial}{\partial y}, -r\sin\theta \frac{\partial}{\partial x} + r\cos\theta \frac{\partial}{\partial y} \right\rangle$$

$$= r^2$$

$$v = v_r \frac{\partial}{\partial r} + v_\theta \frac{\partial}{\partial \theta}$$

$$\|v\|^2 = v_r^2 + r^2 v_\theta^2$$

physicists do this

$$\mathcal{L} = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2)$$

$x = r\cos\theta$
$y = r\sin\theta$

$$\dot{x} = \dot{r}\cos\theta - r\sin\theta\,\dot{\theta}$$

$$\text{- - - -} \Rightarrow \mathcal{L} = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2)$$

Mechanics

q position of a particle $\qquad V(q)$ potential energy

v velocity of particle

$$\mathcal{L} = \frac{1}{2}mv^2 - V(q) \quad \rightarrow \quad \text{EL:} \quad m\dot{v} = -\nabla V(q) = \text{force}$$

for fields

$$\mathcal{L} = \frac{1}{2}m\left(\frac{d}{dt}x\right)^2 - V(x) \qquad \| \quad x = x(t)$$

$\downarrow$ field

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - V(\phi) \qquad \| \quad \phi = \phi(t,x)$$

$$\boxed{\frac{\partial}{\partial x^\mu}\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)} - \frac{\partial\mathcal{L}}{\partial\phi} = 0}$$

$$\mathcal{L} = \frac{1}{2}\partial_\mu \partial^\mu - V(\phi)$$

$$\frac{\partial}{\partial x^\mu}\frac{\partial\mathcal{L}}{\partial\partial_\mu}(\phi, \partial\phi) - \frac{\partial\mathcal{L}}{\partial\phi}(\phi, \partial\phi)$$

in Minkowski space $\partial_t^2 - \nabla^2$

$$\rightarrow \quad \partial_\mu\partial^\mu\phi = -V'(\phi)$$

<u>in QFT</u> start with free Lagrangian for your kind of field

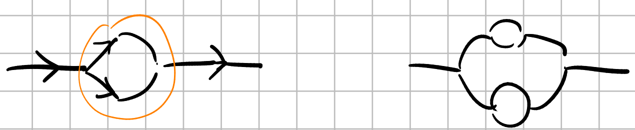
$$\mathcal{L}_F = \frac{1}{2}\left(\partial_\mu\phi\,\partial^\mu\phi - m^2\phi^2\right)$$

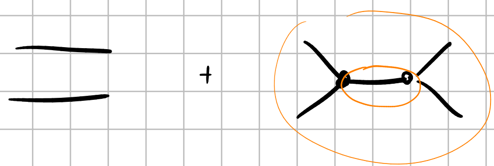
$$\mathcal{L}_T = -\frac{1}{4}\mathrm{tr}\left(F_{\mu\nu}F^{\mu\nu}\right)$$

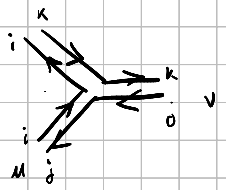
<u>ex:</u> add ϕ^3 interaction: $\mathcal{L} = \frac{1}{2}\left(\partial_\mu\phi\,\partial^\mu\phi - m^2\phi^2\right) + g\,\phi^3$

Canonical quantization $\quad \phi(p) \rightarrow \hat{\phi}(p)$ (operator)

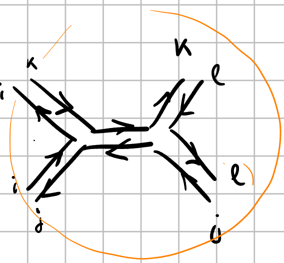
$$\langle 0|\phi(q)\,\phi(p)|0\rangle = \delta(p-q)\,\frac{1}{p^2} \quad \text{(free Lagrangian)}$$



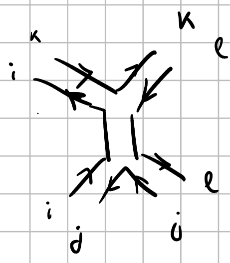




$(A_\mu)^i{}_j$



s chanel

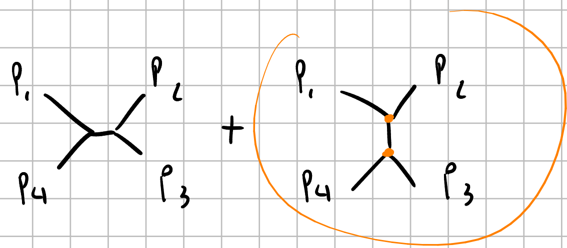


t chanel

planar graphs

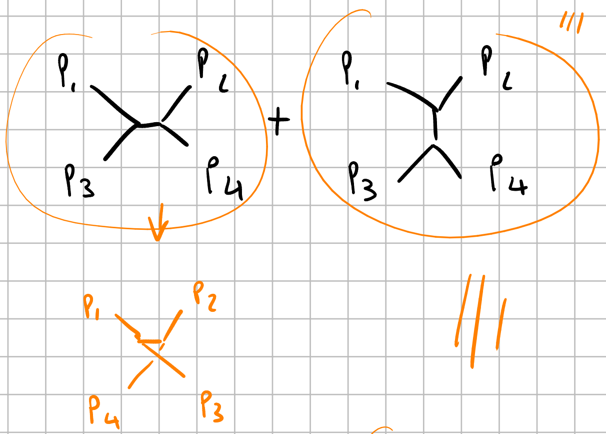
$$A_n = \sum_{\sigma \in S_n/\mathbb{Z}_n} (- \text{——})\, A_n(\sigma(1), \sigma(2), \dots, \sigma(n))$$

$A_4(1,2,3,4) =$



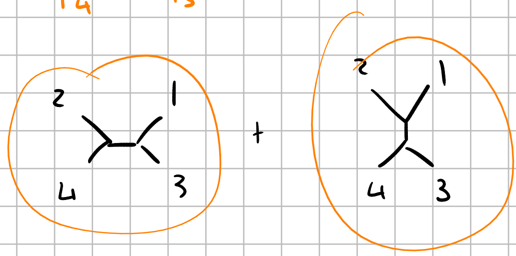
$\frac{1}{(p_1+p_2)^2}$

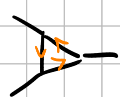
$A_4(1,2,4,3) =$



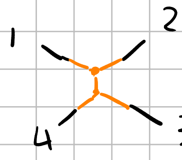
$\frac{1}{(p_1+p_2)^2}$

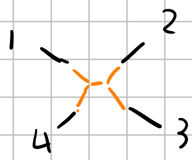
$A_4(2\,1\,3\,4) =$



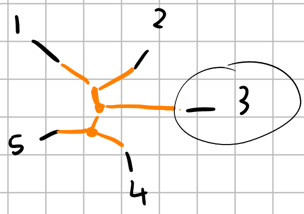
- tree-level diagrams: no loops  not allowed!
- fixed ordering
- planar diagrams
- conected

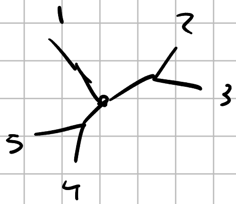
$n = 4$

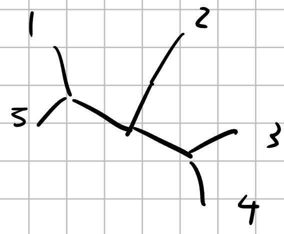




$n = 5$

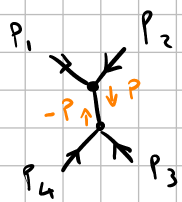




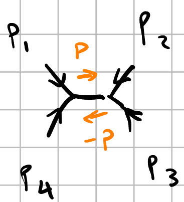


<u>rule</u> each internal edge is a propagator $\frac{1}{p^2}$

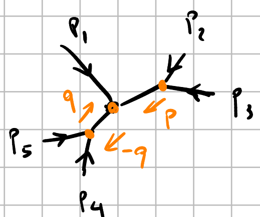
at each vertex we have conservation of momentum



$$\frac{1}{p^2} = \frac{1}{(P_1+P_2)^2} = \frac{1}{(-P_3-P_4)^2}$$

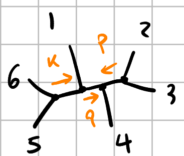


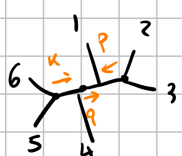
$$\frac{1}{p^2} = \frac{1}{(P_1+P_4)^2} = \frac{1}{(P_2+P_3)^2}$$



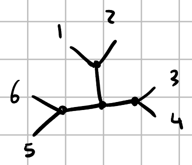
$$\frac{1}{(P_2+P_3)^2} \quad \frac{1}{(P_4+P_5)^2}$$

$$-q = P_1 + P = P_1 + P_2 + P_3 = -P_4 - P_5$$



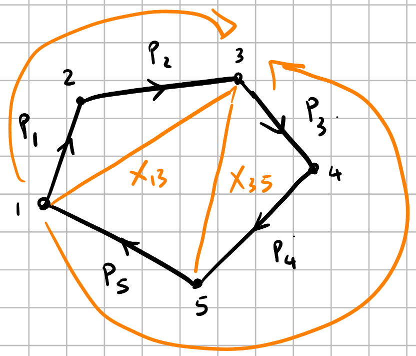


$$\underset{X_{24}}{\frac{1}{(P_2+P_3)^2}} \cdot \underset{X_{25}}{\frac{1}{(P_2+P_3+P_4)^2}} \cdot \underset{X_{51}}{\frac{1}{(P_5+P_6)^2}}$$



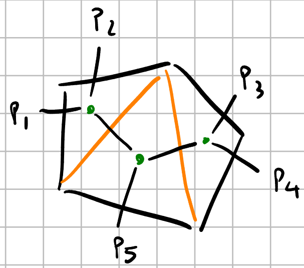
$$\frac{1}{(p_1+p_2)^2}\cdot\frac{1}{(p_3+p_4)^2}\cdot\frac{1}{(p_5+p_6)^2}$$

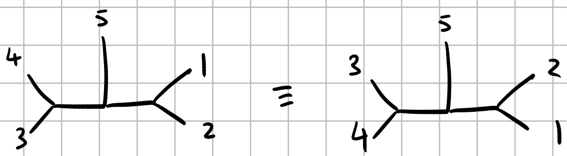
X_{13} $\quad$ X_{35} $\quad$ X_{51}



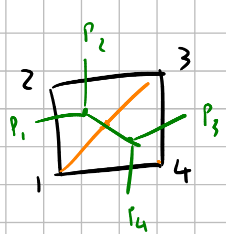
$\sum_i p_i = 0$

$$\frac{1}{X_{13}}\,\frac{1}{X_{35}} = \frac{1}{(p_1+p_2)^2}\,\frac{1}{(p_3+p_4)^2}$$

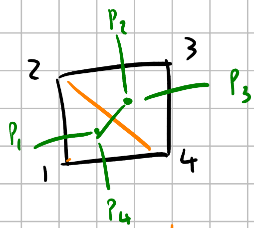




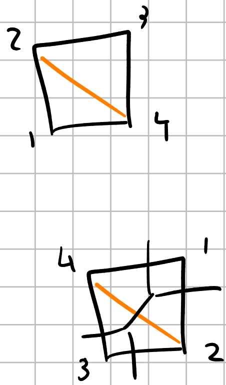
n = 4



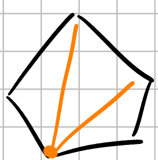
$\frac{1}{X_{13}}$



$\frac{1}{X_{24}}$

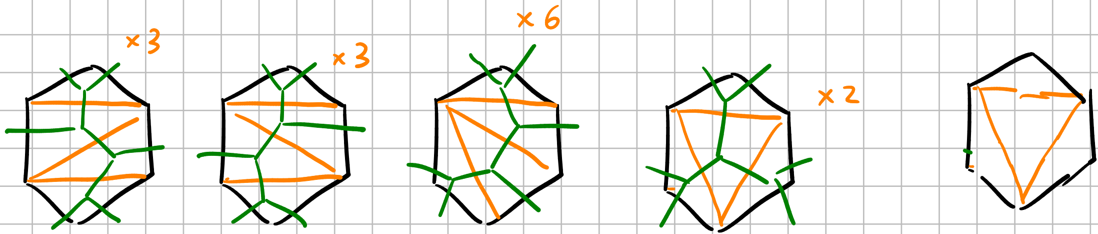


n = 5



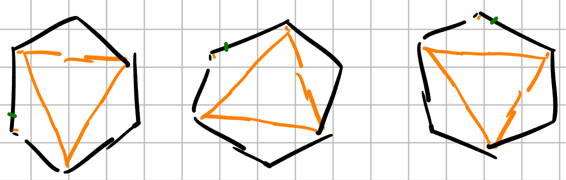
+ cyclic

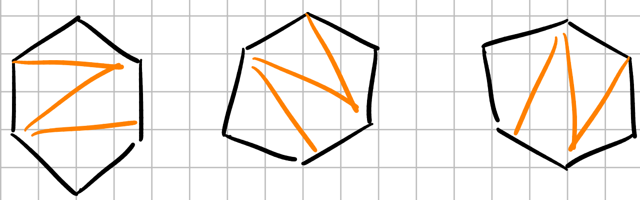
n = 6

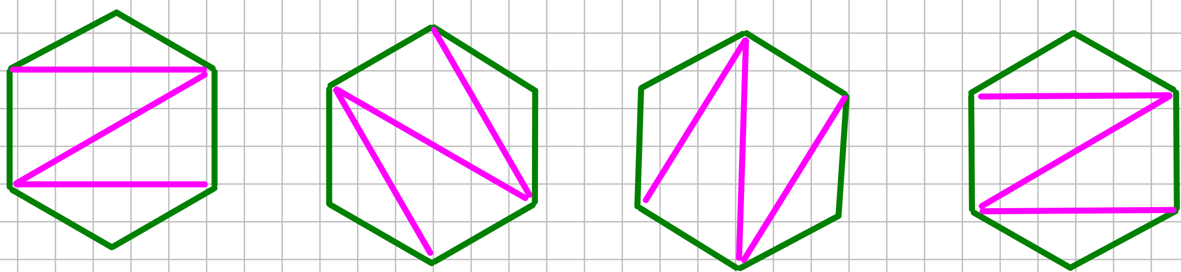


number of planar diagrams with fixed ordering is the Catalan number

C_k $\qquad C_0 = 1 \qquad C_1 = 1 \qquad C_2 = 2 \qquad C_3 = 5 \qquad C_4 = 14$



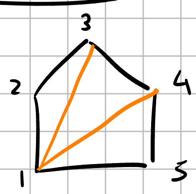




$$\frac{1}{n+1}\binom{2n}{n}$$

---

triangulation $\mathcal{T}$



pick X_{13} X_{14} as planar

everything but $c_{i-1,j-1}$ with $(i,j) \in \{(1,3),(1,4)\}$

exclude c_{52} c_{53} and c_{15} c_{45}

and c_{12} c_{23} c_{34} → because planar

$n = 5$

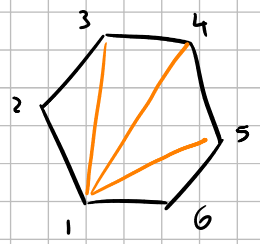
number of invariants: $\frac{n(n-3)}{2} = 5$

c_{12}	c_{13}	c_{14}	$\cancel{c_{15}}$
	c_{23}	c_{24}	$\cancel{c_{25}}$
		c_{34}	$\cancel{c_{35}}$
			$\cancel{c_{45}}$

$$c_{12} = -X_{13} = -(P_1+P_2)^2$$

planar: sum of consecutive momenta squared

$$c_{ij} = -(P_i+P_j)^2$$



take X_{13} X_{14} X_{15} | exclude c_{62} c_{63} c_{64}

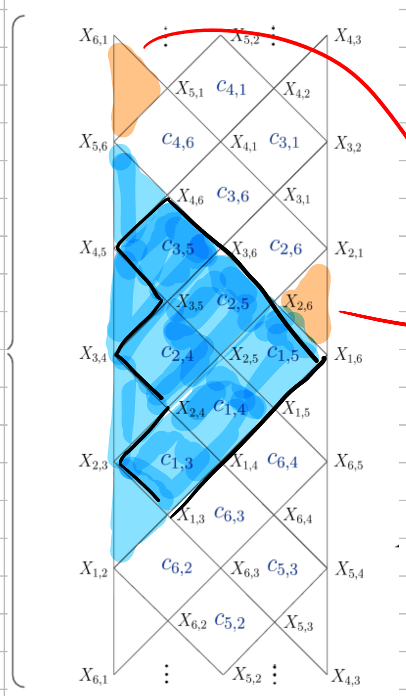
$c_{i-1,j-1}$

c_{12}	c_{13}	c_{14}	c_{15}	c_{16}
	c_{23}	c_{24}	c_{25}	$\cancel{c_{26}}$
		c_{34}	c_{35}	$\cancel{c_{36}}$
			c_{45}	$\cancel{c_{46}}$
				c_{56}

$$c_{ij} = -(p_i+p_j)^2$$

replace

X_{24} with c_{13}
X_{25} with c_{24}
X_{26} with c_{15}
X_{35} c_{24}
X_{36} c_{25}
X_{46} c_{35}



planar $c_{16} = -X_{62}$

$n=6$: 9 independent variables

---

in the $n=5$ example: X_{13} X_{14} $\cancel{c_{12}}$ c_{13} c_{14} $\cancel{c_{23}}$ c_{24} $\cancel{c_{34}}$

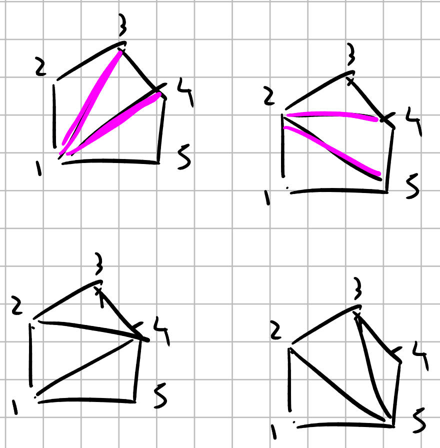
=

non planar

want to show that the remaining ones (X_{24} X_{25} X_{35})

are obtained from basis X_{13} X_{14}

$X_{12} = p_1^2 = 0$



→ these pictures are to find the non-zero X_{ij}s

$$c_{ij} = X_{ij} + X_{i+1\, j+1} - X_{i\, j+1} - X_{i+1\, j}$$

$$c_{13} = X_{13} + X_{24} - X_{14}$$

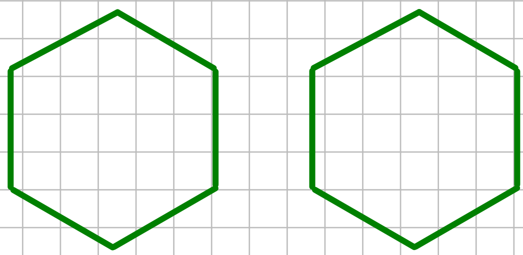
$$c_{14} = X_{14} + X_{25} - X_{24}$$

$$c_{24} = X_{24} + X_{35} - X_{25}$$

$$X_{25} = c_{14} + c_{13} - X_{13}$$

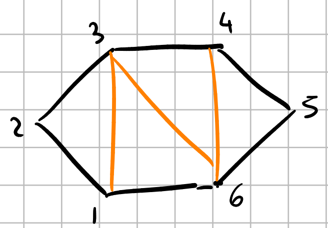
$$X_{35} = c_{14} + c_{24} - X_{14}$$

$$X_{24} = c_{24} - X_{35} + X_{25}$$



Procedure to replace variables:

- choose triangulation, keep the chords
- for each discarded X_{ij}, replace with $c_{i-1,\, j-1}$



X_{13} X_{36} X_{46}

replace:

$X_{14} \to c_{63}$

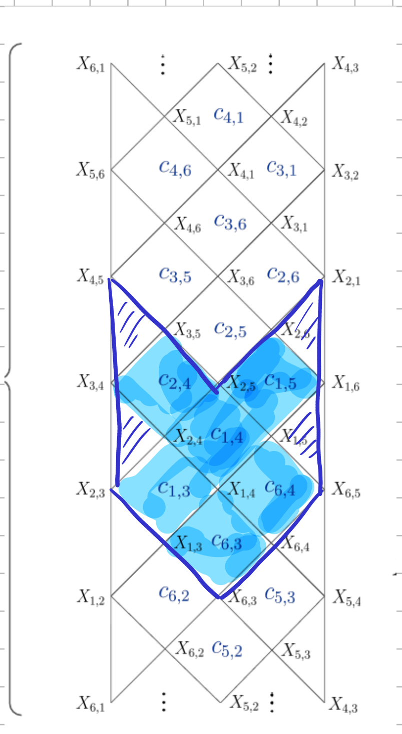
$X_{15} \to c_{64}$

$X_{24} \to c_{13}$

$X_{25} \to c_{14}$

$X_{26} \to c_{15}$

$X_{35} \to c_{24}$



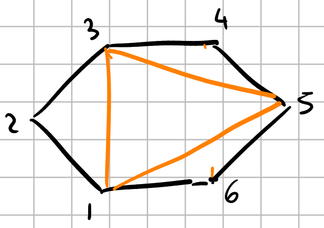
$$c_{23} = \overset{0}{X_{23}} + \overset{0}{X_{34}} - X_{24} - \overset{0}{X_{33}}$$

$$= -X_{24}$$

if $i \leq j$ $\quad X_{ij} = \left(\sum_{k=i}^{j-1} p_k\right)^2$

$$X_{i\, i+1} = p_i^2 = 0$$

$X_{ii} = 0$ by definition



X_{13} X_{35} X_{51}

replace:

$X_{14} \to c_{63}$

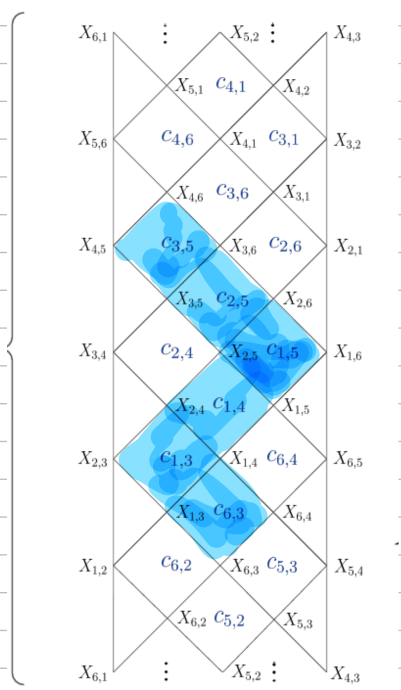
$X_{24} \to c_{13}$

$X_{25} \to c_{14}$

$X_{26} \to c_{15}$

$X_{36} \to c_{25}$

$X_{46} \to c_{35}$



$$\mathcal{L} = \frac{1}{2} tr\left(\partial_\mu \phi \, \partial^\mu \phi\right) + g\, tr\left(\phi^3\right)$$

ϕ is an $N \times N$ matrix

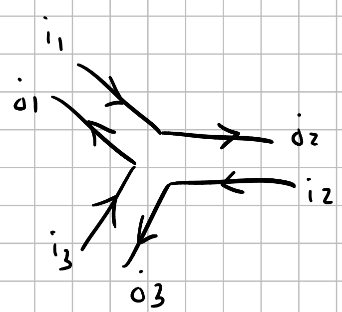
$\phi^{i}{}_{j}$ ← Colour indices

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi^{i}{}_{j} \, \partial^\mu \phi^{j}{}_{i} + g\, \phi^{i}{}_{j} \phi^{j}{}_{k} \phi^{k}{}_{i}$$

$$\langle \phi^{i}{}_{j}(x)\, \phi^{k}{}_{e}(y) \rangle = \frac{\int D\phi \; e^{iS[\phi]} \; \phi^{i}{}_{j}(x)\, \phi^{k}{}_{e}(y)}{\int D\phi \; e^{iS[\phi]}}$$

$$\langle \phi^{i_1}{}_{j_1}(x_1)\, \phi^{i_2}{}_{j_2}(x_2) \rangle_0 \propto \delta^{i_1}_{j_2}\, \delta^{i_2}_{j_1}$$

free



$$\langle \varphi^{i_1}{}_{j_1}(p_1)\, \varphi^{i_2}{}_{j_2}(p_2)\, \varphi^{i_3}{}_{j_3}(p_3) \rangle$$

$$e^{S[\phi]} = e^{S_0[\phi]} \left(1 + g \int tr(\phi^3) + \frac{g^2}{2} \left(\int tr(\phi^3) \right)^2 + \dots \right)$$

$$\langle \varphi^{i_1}{}_{j_1}(p_1)\,\varphi^{i_2}{}_{j_2}(p_2)\,\varphi^{i_3}{}_{j_3}(p_3)\rangle$$

$$\parallel$$

expansion of $e^{g\int tr\,\phi^3}$

$$\approx \frac{\int D\varphi\, e^{-S_0[\varphi]}\left(1+g\int tr(\phi^3)\right)\varphi^{i_1}{}_{j_1}(p_1)\,\varphi^{i_2}{}_{j_2}(p_2)\,\varphi^{i_3}{}_{j_3}(p_3)}{\int D\varphi\, e^{-S_0[\varphi]}}$$

$\phi^{i_4}{}_{j_4}\;\phi^{j_4\,i_5}{}_{j_5}\;\phi^{j_5\,i_6}{}_{i_4\,j_6}$

$\varphi^{i_4}{}_{j_4}\,\varphi^{i_5}{}_{j_5}\,\varphi^{i_6}{}_{j_6}\;\delta^{j_6}{}_{i_4}\,\delta^{j_4}{}_{i_5}\,\delta^{j_5}{}_{i_6}$

$$\langle \varphi^{i_1}{}_{j_1}(p_1)\,\varphi^{i_2}{}_{j_2}(p_2)\,\varphi^{i_3}{}_{j_3}(p_3)\rangle \approx$$

$$\langle\, \varphi^{i_1}{}_{j_1}(p_1)\,\varphi^{i_2}{}_{j_2}(p_2)\,\varphi^{i_3}{}_{j_3}(p_3)\int dp_4\,dp_5\,dp_6\;tr\left(\phi(p_4)\phi(p_5)\phi(p_6)\right)\;\delta(p_4+p_5+p_6)\,\rangle_0$$

$\varphi^{i_1}{}_{j_1}(p_1)$

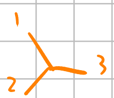
$\to \langle 1\,2\;\;3\,4\;\;5\,6\rangle$

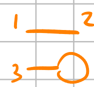
$\langle 1\;2\;3\;4\;5\;6\rangle =$

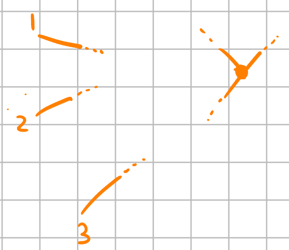
$\langle 1\,2\rangle\langle 3\,4\rangle\langle 5\,6\rangle$
$\langle 1\,2\rangle\langle 3\,5\rangle\langle 4\,6\rangle$
$\langle 1\,3\rangle$ — —

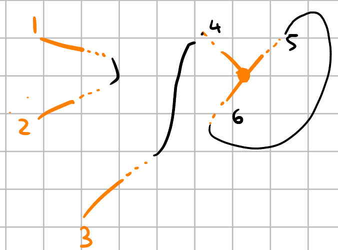
$(6-1)!!$

15 total



 $\to$ Contract 4 5 6 among themselves





$\langle 1\,4\rangle\langle 2\,5\rangle\langle 3\,6\rangle \quad \delta^{j_6}{}_{i_4}\,\delta^{j_4}{}_{i_5}\,\delta^{j_5}{}_{i_6}$

$\langle 1\,4\rangle\langle 2\,6\rangle\langle 3\,5\rangle$

~~⟨1 5⟩⟨2 4⟩⟨3 6⟩~~

~~⟨1 5⟩⟨2 6⟩⟨3 4⟩~~

~~⟨1 6⟩⟨2 4⟩⟨3 5⟩~~

~~⟨1 6⟩⟨2 5⟩⟨3 4⟩~~

double counting

$tr(AB) = \sum_{ij} A^i{}_j B^j{}_i$ $\qquad (AB)^i{}_j = \sum_k A^i{}_k B^k{}_j$

$tr(ABC) = \sum_{ijk} A^i{}_j B^j{}_k C^k{}_i$

A 1 B 1 C

$$= \sum_{\substack{i_1 j_1 \\ i_2 j_2 \\ i_3 j_3}} A^{i_1}{}_{j_1} B^{i_2}{}_{j_2} C^{i_3}{}_{j_3} \delta^{j_1}_{i_2} \delta^{j_2}_{i_3} \delta^{j_3}_{i_1}$$

momentum

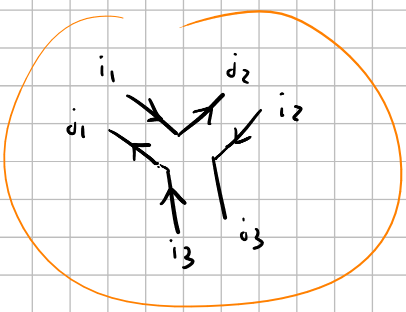
$$\sum_{\substack{i_4 j_4 \\ i_5 j_5 \\ i_6 j_6}} \int dp_4\, dp_5\, dp_6 \underbrace{\langle 1\, 4\rangle}_{\delta^{i_1}_{j_4} \delta^{j_1}_{i_4} \delta(p_1+p_4)} \langle 2\, 5\rangle \langle 3\, 6\rangle\, \delta(p_4+p_5+p_6)\, \delta^{j_6}_{i_4} \delta^{j_4}_{i_5} \delta^{j_5}_{i_6} \leftarrow \text{glue}$$

$$\sum_{i_4} \delta^{j_1}_{i_4} \delta^{j_6}_{i_4} = \delta^{j_1 j_6}$$

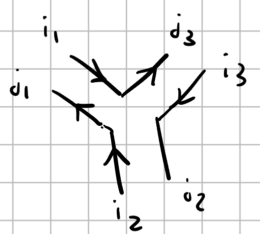
$$\sum_{\substack{i_4 j_4 \\ i_5 j_5 \\ i_6 j_6}} \delta^{i_1}_{j_4} \delta^{j_1}_{i_4} \delta^{i_2}_{j_5} \delta^{j_2}_{i_5} \delta^{i_3}_{j_6} \delta^{j_3}_{i_6} \delta^{j_6}_{i_4} \delta^{j_4}_{i_5} \delta^{j_5}_{i_6} = \delta^{i_1}_{j_2} \delta^{i_2}_{j_3} \delta^{i_3}_{j_1}$$

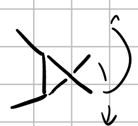
$$\|$$

$$\sum_{\substack{j_4 \\ i_5 j_5 \\ i_6 j_6}} \delta^{i_1}_{j_4} \delta^{i_2}_{j_5} \delta^{j_2}_{i_5} \delta^{i_3}_{j_6} \delta^{j_3}_{i_6} \delta^{j_1 j_6} \delta^{j_4}_{i_5} \delta^{j_5}_{i_6} = \sum_{\substack{j_4 \\ i_5 j_5 \\ i_6}} \delta^{i_3 j_1} \delta^{i_1}_{j_4} \delta^{i_2}_{j_5} \delta^{j_2}_{i_5} \delta^{j_3}_{i_6} \delta^{j_4}_{i_5} \delta^{j_5}_{i_6}$$



$\langle 1\, 4\rangle \langle 2\, 6\rangle \langle 3\, 5\rangle$ non cyclic permutation of previous one swapping 2 and 3

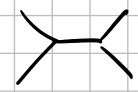
$\rightarrow$ 



$\mathcal{A}_3 = \sum_{S_n/\mathbb{Z}_n}$ of amplitude with fixed ordering

---

n = 4 $\quad \langle \underbrace{1\,2\,3\,4}_{\text{external legs}}\ \underbrace{5\,6\,7}_{tr(\phi^3)}\ \underbrace{8\,9\,10}_{tr(\phi^3)} \rangle$



cyclic symmetry $\overrightarrow{567}$ $\overrightarrow{8\,9\,10}$

this time if we want connected tree level things, we need to connect one of 567 with one of 8 9 10

fix connection between 7 and 10

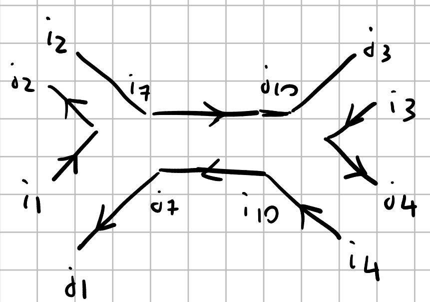
the rest needs to be arranged without loops

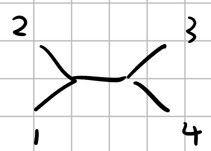
$\langle 1\,5\rangle\langle 2\,6\rangle\langle 3\,8\rangle\langle 4\,9\rangle\langle 7\,10\rangle$ • permutation of 1 2 3 4

$\delta^{i_4}_{j_9}\,\delta^{i_9}_{j_4}$

$\langle 1\,5\rangle\langle 2\,6\rangle\langle 3\,8\rangle\langle 4\,9\rangle\langle 7\,10\rangle\ \delta^{j_7}_{i_5}\,\delta^{j_5}_{i_6}\,\delta^{j_6}_{i_7}\quad \delta^{j_{10}}_{i_8}\,\delta^{j_8}_{i_9}\,\delta^{j_9}_{i_{10}}$

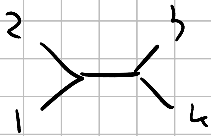
$\delta^{i_1}_{j_5}\,\delta^{i_5}_{j_1}\qquad \delta^{i_2}_{j_6}\,\delta^{i_6}_{j_3}\qquad \delta^{i_7}_{j_{10}}\,\delta^{i_{10}}_{j_7}$



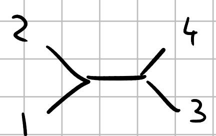


fixed colour ordering

w/o colours



$\equiv$

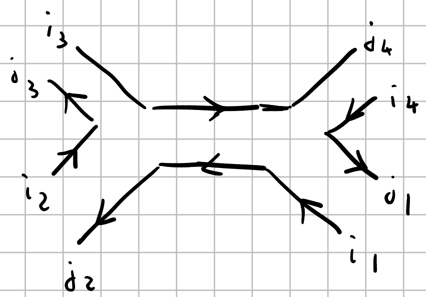


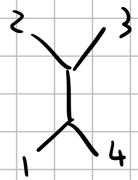
$\langle 1\,5\rangle\langle 2\,6\rangle\,\langle 3\,8\rangle\langle 4\,9\rangle\langle 7\,10\rangle$

$\downarrow$ cyclic permutation

$\langle 2\,5\rangle\langle 3\,6\rangle\langle 4\,8\rangle\,\langle 1\,9\rangle\,\langle 7\,10\rangle$

other diagrams



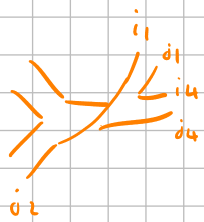


$1 \to 2$

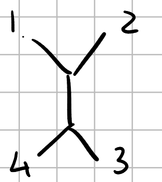
$2 \to 3$

$3 \to 4$

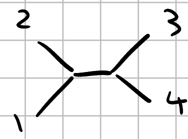
$4 \to 1$



$\langle 3\,5\rangle\langle 4\,6\rangle\langle 1\,8\rangle\,\langle 2\,9\rangle\,\langle 7\,10\rangle$



$\equiv$



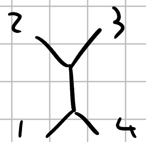
in the paper $\mathrm{Tr}(T^{\sigma(1)}\dots T^{\sigma(n)})$

$$\mathcal{A}_4 = \sum_{\sigma\in S_4/\mathbb{Z}_4} \delta^{i_1}_{j_2}\,\delta^{i_2}_{j_3}\,\delta^{i_3}_{j_4}\,\delta^{i_4}_{j_1}\; \underbrace{A_4(\sigma(1)\dots\sigma(4))}$$

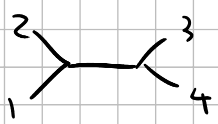
monoton amplitudes (planar)

(partial amplitude)

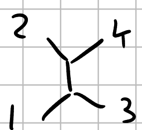
$A_4(1\,2\,3\,4) =$



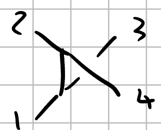
$+$



this includes

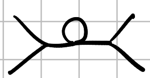


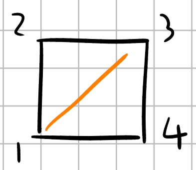
$\equiv$

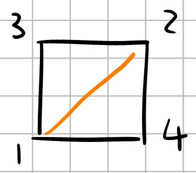


(u- channel)

factor of N




2
3
1
4


3
2
1
4

↙ partial ordered amplitude

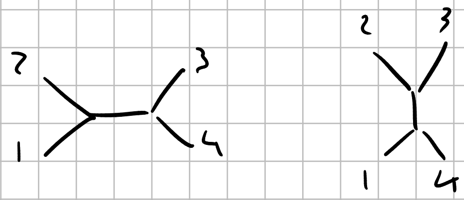
$\mathrm{Res}\, A_n(1\,2\,3 \ldots n)$ w.r.t one of the X_{ij} (the poles)

factorizes as product of two smaller amplitudes

if $f(z)$ is a complex function with Laurent series

$$= \sum_{n\in\mathbb{Z}} a_n z^n$$

$$\mathrm{Res}\, f = a_{-1}$$



$$A_4(1234) \propto \frac{1}{s_{12}} + \frac{1}{s_{23}}$$

look at residue near one of the poles (ex: s_{12})

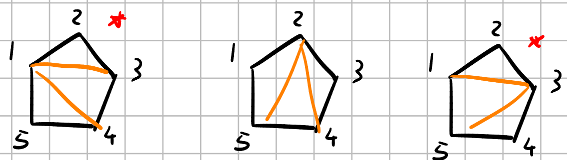
$$\mathrm{Res}\,(A_4)_{s_{12}} = 1$$

$$\propto \int dp\; A_3(1\,2\,p)\, A_3(-p\,3\,4)$$

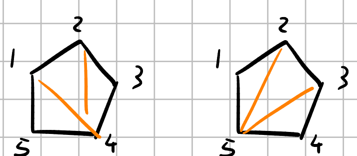
$$A_4 = \frac{1}{s_{12}} \boxed{A_{\text{left}}\; A_{\text{right}}} + O(1)$$

---

if we want to expand at $s_{12} = X_{13}$



$$A_5(1\,2\,3\,4\,5) = \boxed{\frac{1}{X_{13}X_{14}} + \frac{1}{X_{13}X_{35}}} + \frac{1}{X_{24}X_{25}} + \frac{1}{X_{24}X_{14}} + \frac{1}{X_{25}X_{35}}$$

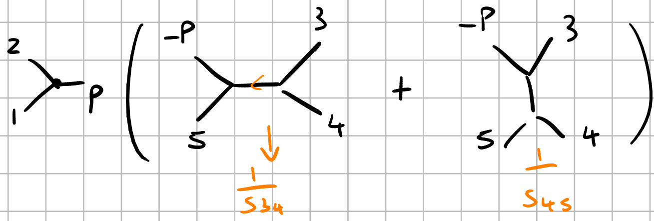


$$\mathrm{Res}\,(A_5)_{X_{13}} = \frac{1}{X_{14}} + \frac{1}{X_{35}} =$$

X_{14} ↙ s_{45} $\quad$ X_{35}: s_{34}

$(p_1+p_2+p_3)^2$

$= (p_4+p_5)^2$



$$= A_3(1\,2\,p)\,A_4(-p\,3\,4\,5)$$

this follows from $S^\dagger S = \mathbb{1}$ (S - matrix)

tree amplitude A_n

choose 2 momenta i, j

$$p_i \to p_i + zq \qquad z \in \mathbb{C}$$
$$p_j \to p_j - zq$$

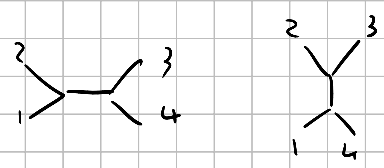
q is a random vector s.t. $p_i \cdot q = p_j \cdot q = q^2 = 0$

this does not effect that $p_i^2 = p_j^2 = 0$ and does not change s_{ij}

$$(p_i + zq)^2 = p_i^2 + 2z\,\underbrace{q \cdot p_i}_{\to 0} + z^2 \underbrace{q^2}_{\to 0} = p_i^2$$

the new amplitude depends on z

$$A(0) = \oint dz\,\frac{A(z)}{z}$$



ex:

$$A_4(1\,2\,3\,4) = \frac{1}{s_{12}} + \frac{1}{s_{23}}$$

$$p_1 \to p_1 + qz$$
$$p_2 \to p_2 - qz$$

$$s_{12} \to s_{12} = (p_1 + p_2)^2$$
$$s_{23} \to (p_2 - qz + p_3)^2$$

$$A_4(1234)(z) = \frac{1}{(p_1 + p_2)^2} + \frac{1}{(p_2 - qz + p_3)^2}$$

$$\frac{A(z)}{z} = \frac{A(0)}{z} + A'(0) + \frac{A''(0)}{2} z + \dots$$

$A_4(0)$ = original amplitude

$\frac{A(z)}{z}$ pole at $z = 0$ expand in a series: $\mathrm{Res}\,\frac{A(z)}{z} = A(0)$

$$A(0) = \mathrm{Res}\, \frac{A(z)}{z} \propto \oint \frac{A(z)}{z}\, dz$$

$s_{14} = (P_1 + P_4)^2 =$

look at $A_4(1234) = \frac{1}{s_{12}} + \frac{1}{s_{14}} = \frac{1}{s_{12}} - \frac{1}{s_{12} + s_{13}}$

$$s_{12} + s_{13} = (P_1 + P_2)^2 + (P_1 + P_3)^2 = 2P_1 \cdot P_2 + 2P_1 \cdot P_3 = 2P_1 \cdot (P_2 + P_3)$$

$$= 2P_1(P_1 + P_2 + P_3) = 2P_1 \cdot (-P_4) = -(P_1 + P_4)^2 = -s_{14}$$

if $s_{13} = 0$ then $A_4(1234) = 0$

$$A_4 = \frac{1}{2P_1 \cdot P_2} + \frac{1}{2P_1 \cdot P_4}$$

$P_1 \cdot P_3 = 0$

$P_1 \cdot (P_1 + P_2 + P_3 + P_4) = 0$

$\downarrow$

$P_1 \cdot P_2 = -P_1 \cdot P_4$

$$A_5 = \frac{1}{x_{13}x_{14}} + \frac{1}{x_{13}x_{35}} + \frac{1}{x_{24}x_{25}} + \frac{1}{x_{24}x_{14}} + \frac{1}{x_{25}x_{35}}$$

$$= \frac{1}{s_{12}\, s_{45}} + \frac{1}{s_{23}\, s_{15}} + \frac{1}{s_{34}\, s_{21}} + \frac{1}{s_{45}\, s_{32}} + \frac{1}{s_{51}\, s_{43}}$$

$P_1 + P_2 + P_3 + P_4 + P_5 = 0$ $\quad$ $P_i^2 = 0$

$s_{ij} = 2P_i \cdot P_j$

$$\frac{1}{s_{12}}\left(\frac{1}{s_{45}} + \frac{1}{s_{34}}\right) + \frac{1}{s_{23}}\left(\frac{1}{s_{15}} + \frac{1}{s_{45}}\right) = -\frac{1}{s_{15}\, s_{43}}$$

$$\frac{1}{s_{12}}\left(\frac{s_{15}s_{43}}{s_{45}} + s_{15}\right) + \frac{1}{s_{23}}\left(s_{43} + \frac{s_{15}s_{43}}{s_{45}}\right) = -1$$

$s_{43}=0$ $\qquad\qquad \dfrac{s_{15}}{s_{12}} = -1$

$$s_{13} = s_{14} = 0$$

$$\frac{1}{s_{12}\,s_{45}} + \frac{1}{s_{23}\,s_{15}} + \frac{1}{s_{34}\,s_{21}} + \frac{1}{s_{45}\,s_{32}} + \frac{1}{s_{51}\,s_{43}}$$

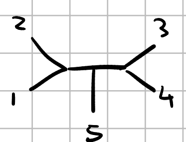
why	13	14
why	24	25
	31	35
	42	41
	52	53

$p_1\cdot p_3 = 0 \qquad p_1\cdot p_4 = 0 \qquad\qquad p_1\cdot p_2 = -p_1\cdot p_5$

$p_1\cdot(p_2+p_3+p_4\cdot p_5) = 0 \qquad\qquad s15 = -s_{12}$

$$\cancel{\frac{1}{s_{12}\,s_{45}}} - \frac{1}{s_{23}\,s_{12}} + \cancel{\frac{1}{s_{34}\,s_{12}}} + \frac{1}{s_{45}\,s_{32}} - \cancel{\frac{1}{s_{12}\,s_{43}}}$$

$$\frac{1}{s_{12}\,s_{34}}$$

$A_5 =$  $+$ cyclic

$$= \frac{1}{s_{12}\,s_{34}} + \frac{1}{s_{23}\,s_{45}} + \frac{1}{s_{34}\,s_{51}} + \frac{1}{s_{45}\,s_{12}} + \frac{1}{s_{51}\,s_{23}}$$

$p_1\cdot p_3 = 0 \qquad p_1\cdot p_4 = 0 \qquad\qquad p_1\cdot p_2 = -p_1\cdot p_5$

$p_1\cdot(p_2+p_3+p_4\cdot p_5) = 0 \qquad\qquad s15 = -s_{12}$

$p_1\cdot p_2 + p_1\cdot p_5 = 0$

$$\cancel{\frac{1}{s_{12}\, s_{34}}} + \frac{1}{s_{23}\, s_{45}} - \cancel{\frac{1}{s_{34}\, s_{12}}} + \frac{1}{s_{45}\, s_{12}} - \frac{1}{s_{12}\, s_{23}}$$

$$\frac{1}{s_{23}\, s_{45}} + \frac{1}{s_{45}\, s_{12}} - \frac{1}{s_{12}\, s_{23}}$$

$$1 + \frac{s_{23}}{s_{12}} - \frac{s_{45}}{s_{12}} \Rightarrow \frac{\boxed{s_{12} + s_{23} - s_{45}}}{s_{23}\, s_{12}\, s_{45}} = 0$$

$$p_2 \cdot (p_1 + p_3 + p_4 \cdot p_5) = 0$$

$$s_{12} + \cancel{s_{23}} + \cancel{s_{24}} + \cancel{s_{25}} = 0$$

$$p_i \Big(\sum_j p_j \Big) = 0 \Rightarrow \sum_j s_{ij} = 0$$

$$i=2 \qquad \underline{s_{21}} + \underline{s_{23}} + \cancel{s_{24}} + \cancel{s_{25}} = 0 \qquad \oplus$$

$$i=4 \qquad \cancel{s_{42}} + \cancel{s_{43}} + \underline{\underline{s_{45}}} = 0 \qquad \ominus$$

$$i=3 \qquad \underline{s_{32}} + \cancel{s_{34}} + \cancel{s_{35}} = 0 \qquad \oplus$$

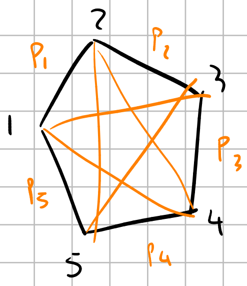
$$i=5 \qquad -\underline{\underline{s_{12}}} + \cancel{s_{52}} + \cancel{s_{53}} + \underline{\underline{s_{54}}} = 0 \qquad \ominus$$

$$\underline{\underline{s_{12}}} + \underline{\underline{s_{23}}} - \underline{\underline{s_{45}}}$$

$$2 \left(s_{21} + s_{23} - s_{45} \right) = 0$$

$$s_{13} = s_{14} = 0$$

$$s_{24} = s_{25} = 0$$



X_{13}

X_{14}

X_{24}

X_{25}

X_{35}

$$c_{ij} = -s_{ij} = -(p_i + p_j)^2$$

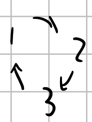
$$X_{13} = (p_1 + p_2)^2$$

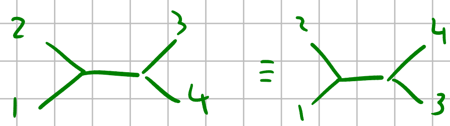
↳ planar

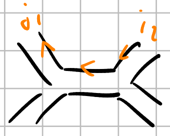
$$X_{25} = (p_2 + p_3 + p_4)^2$$

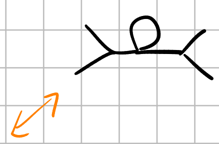
$c_{12} = X_{13}$ planar

$c_{13} = (p_1 + p_3)^2$ not planar if $n > 3$

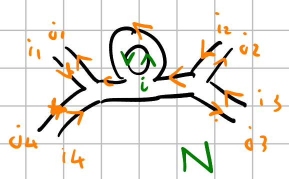


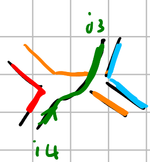






$$\langle \phi^{i_1}_{j_1}(p_1)\, \phi^{i_2}_{j_2}(p_2) \ldots \rangle$$

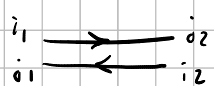


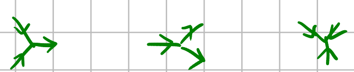


$$\mathcal{L} = \frac{1}{2} \mathrm{tr}(\underbrace{\partial_\mu \phi\, \partial^\mu \phi}) + V(\phi)$$

↳ gives propagator

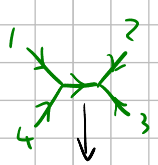
$$\sum_{\substack{i_1 j_1 \\ i_2 j_2}} \partial_\mu \phi^{i_1}_{j_1}\, \partial^\mu \phi^{i_2}_{j_2}\, \delta^{j_1}_{i_2}\, \delta^{i_1}_{j_2}$$



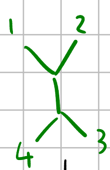


1 2

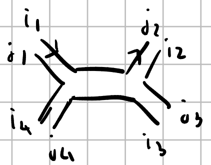
4 3

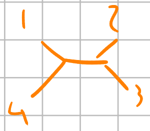


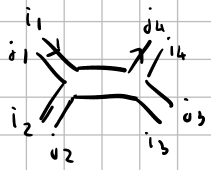
$$\frac{1}{(p_1 + p_4)^2}$$



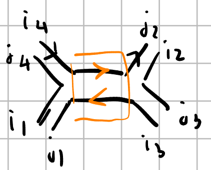
$$\frac{1}{(p_1 + p_2)^2}$$

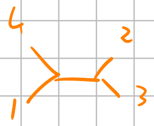




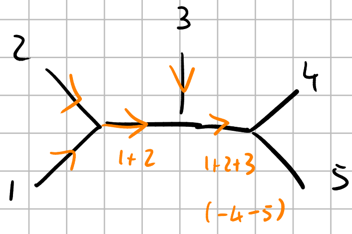


$\frac{1}{(P_1+P_4)^2}\,\delta^{i_1}_{\bar{\imath}_2}\,\delta^{i_2}_{\bar{\imath}_3}\dots$ $\qquad$ $\frac{1}{(P_1+P_2)^2}\,\delta^{i_1}_{\bar{\imath}_4}$





$\frac{1}{(P_1+P_4)^2}\,\delta^{i_1}_{\bar{\imath}_4}$

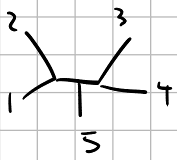


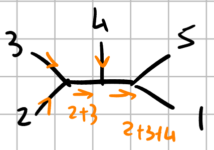
$\frac{1}{X_{13}}\;\frac{1}{X_{14}}$

s_{12} $\quad$ s_{123}

$$A_5(1,2345) = \frac{1}{X_{13}X_{14}} + \text{cyclic}$$

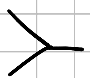
$$\frac{1}{X_{24}X_{25}} + \frac{1}{X_{35}X_{31}} + \dots$$





X_{24} $\quad$ X_{25}

$\frac{1}{s_{12}}\;\frac{1}{s_{12}}$



$p \cdot p$

$p^2 = 0 \;\Rightarrow\; E^2 = \vec{p}^{\,2} \leftarrow$ energy of a photon

Minkowski space: $\quad p_\mu p^\mu = 0 \qquad p_t p_t - p_x p_x - p_y p_y - p_z p_z = 0$

$p_t = E$ energy

$\vec{p} = (p_x\, p_y\, p_z)$ momentum

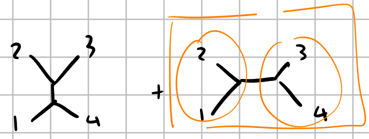
$$A_4 \propto \frac{1}{s_{12}} + \frac{1}{s_{13}} + \frac{1}{s_{14}} =$$

$$\frac{1}{s_{13}}\, A_{\text{left}}\, A_{\text{right}}$$

$$f(z) = \boxed{\frac{a_{-1}}{z} + a_0} + a_1 z + a_2 z^2 + \dots$$

$$\text{Res}\, f(z) = a_{-1}$$

$A_4 =$



$S^\dagger = S$

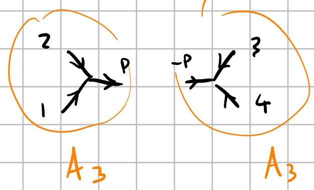
$\Rightarrow$ residue factorization

$$= \frac{1}{s_{14}} + \frac{1}{s_{12}}$$

$\rightarrow O(1)$ w.r.t. s_{12}

$\rightarrow$ 4-momentum $\quad p^0 = E(\vec{p})$

$\text{Res}_{s_{12}=0} A_4 \propto 1$



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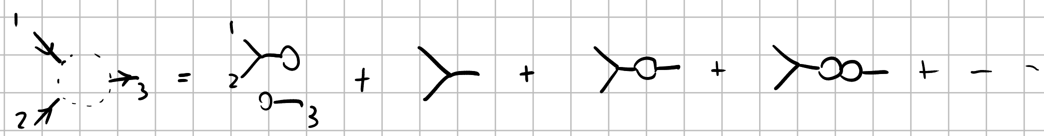
non-local: the QFT cannot be built with path integrals of an action (canonical quantization starting from Lagrangian)

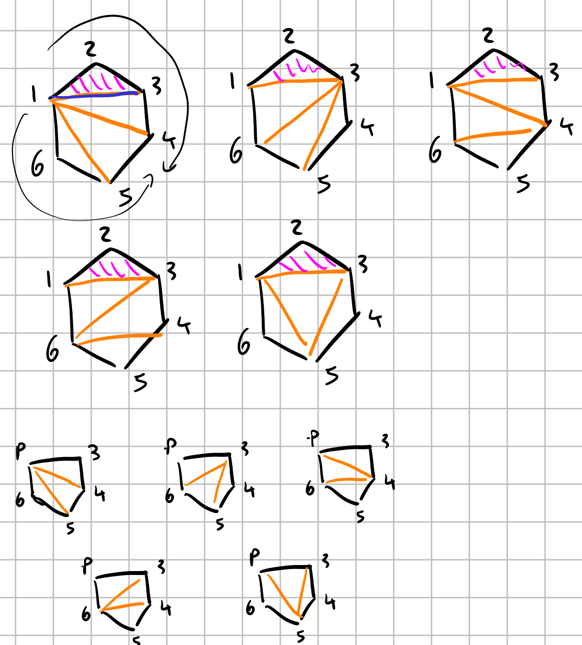


$$f(z,w) = \frac{1}{z} + \frac{1}{w} + \frac{z}{w^2}$$

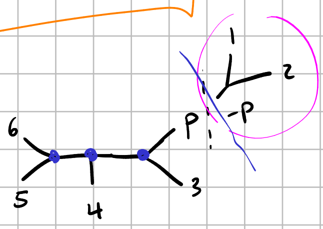
expand in powers of $\frac{1}{z}$ $\qquad f(z,w) = \frac{1}{z} + O(1) \iff \text{Res}\, f_{z=0} = 1$

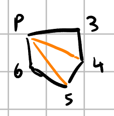
$\text{Res}(A_6)$ look at residue at X_{13}

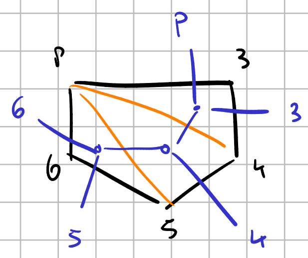
$$A_6 = \frac{1}{X_{13}}\left(\frac{1}{X_{14}X_{15}} + \frac{1}{X_{36}X_{35}} + \frac{1}{X_{14}X_{46}} + \frac{1}{X_{36}X_{46}} + \frac{1}{X_{35}X_{15}} \right) + O(1)$$



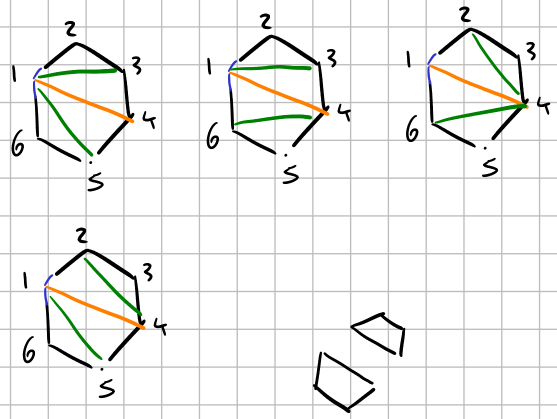
$$\frac{1}{(P_4+P_5+P_6)^2\,(P_5+P_6)^2}$$







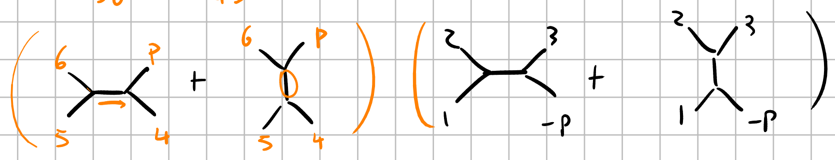
$$A_6 = \frac{1}{X_{14}}\left(\frac{1}{X_{13}X_{15}} + \frac{1}{X_{13}X_{46}} + \frac{1}{X_{24}X_{46}} + \frac{1}{X_{24}X_{15}}\right)$$

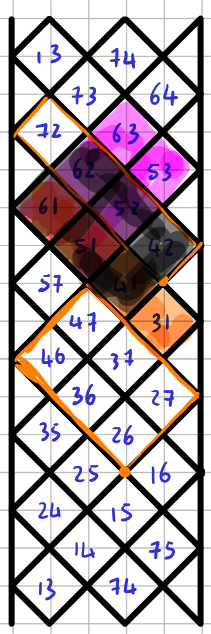


$$= \frac{1}{X_{14}}\left(\frac{1}{X_{15}}\left(\frac{1}{X_{13}} + \frac{1}{X_{24}}\right) + \frac{1}{X_{46}}\left(\frac{1}{X_{13}} + \frac{1}{X_{24}}\right)\right)$$

$$= \frac{1}{X_{14}}\left(\left(\frac{1}{X_{15}} + \frac{1}{X_{46}}\right)\left(\frac{1}{X_{13}} + \frac{1}{X_{24}}\right)\right)$$

$s_{56} + s_{45}$





→ 1 zero

$C_{42} = C_{52} = C_{62} = C_{72} = 0$

→ 2-zero

$C_{26} = C_{36} = C_{46} = 0$
$C_{27} = C_{37} = C_{47} = 0$ $\rightarrow$ $C_{6j} = 0$, $C_{7j} = 0$ $\quad j = 8, \dots, 13$

$C_{14} = C_{15} = C_{16} = 0$

$C_{24} = C_{23} = C_{26} = 0$

1 2 3 4 5 6 7 → 6 7 1 2 3 4 5

$C_{62} = C_{63} = C_{64} = 0$

$C_{72} = C_{73} = C_{74} = 0$

1 zero: $C_{1j} = 0$ $\quad j = 3, \dots, n-1$

2-zero $\begin{cases} C_{1j} = 0 \\ C_{2j} = 0 \end{cases}$ $\quad j = 4, \dots, n-1$

3-zero $\begin{cases} C_{1j} = 0 \\ C_{2j} = 0 \\ C_{3j} = 0 \end{cases}$ $\quad j = 5, \dots, n-1$

$$\boxed{\text{k-zero: } \begin{cases} C_{1j} = 0 \\ C_{2j} = 0 \\ \vdots \\ C_{kj} = 0 \end{cases} \quad j = k+2, \dots, n-1}$$

$\sum_j c_{1j} = 0$ with 1-zero $\Rightarrow$ $s_{12} = -s_{1n}$

$c_{ij} = X_{ij} + X_{i+1\,j+1} - X_{i+1\,j} - X_{i\,j+1}$

$n=7$ $\quad$ $k=3$: $\quad j = 5, 6$

$c_{15} = c_{16} = 0$
$c_{25} = c_{26} = 0$
$\vdots$
$c_{73} = c_{76} = 0$

$1 \le i < j \le n$

$$X_{ij} = \left(\sum_{k=i}^{j-1} p_k\right)^2$$

Q: which X_{ij} are affected by setting $c_{1j}=0$ for $j=3,\dots,n-1$?

$c_{ij} = -(p_i + p_j)^2$

$c_{1j} = -(p_1 + p_j)^2 = -2p_1 \cdot p_j$

$X_{1\,j+1}$ depends on p_1

$\|$

$$\left(\sum_{k=1}^{j} p_k\right)^2 = (\overbrace{p_1 + p_2}^{A} + \overbrace{\cdots + p_j}^{B})^2 \qquad A^2 + 2AB + B^2$$

$$= \underbrace{(p_1+p_2)^2}_{s_{12}} + \underbrace{2(p_1+p_2)\cdot(p_3 + \cdots + p_j)} + \underbrace{(p_3 + \cdots + p_j)^2}_{s_{3,4,\dots j}}$$

$$2(p_1+p_2)\cdot(p_3+\cdots+p_j) = 2p_1\cdot(p_3+\cdots+p_j) + 2p_2(p_3 + \cdots p_j)$$

$$= \sum_{i=3}^{j} \underbrace{2p_1\cdot p_i}_{s_{1i}} + \sum_{i=3}^{j} \underbrace{2p_2\cdot p_i}_{s_{2i}}$$

$$s_{i_1 i_2 \dots i_k} = (p_{i_1} + p_{i_2} + \cdots + p_{i_k})^2$$

$$\int d^3p \; |p\rangle\langle p| = \mathbb{1}$$

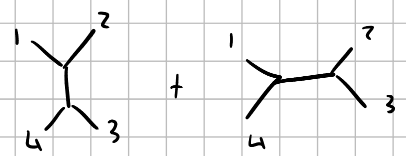
$$S^\dagger = S = \lim_{T\to\infty} U(T,-T)$$

$$S = \mathbb{1} + iT$$

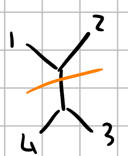
$$S^\dagger S = \mathbb{1} \;\Rightarrow\; \mathbb{1} + iT^\dagger - iT + T^\dagger T = \mathbb{1}$$

$$\langle 0|\, T\, |p_1 p_2 \cdots p_n\rangle = \mathcal{A}_n$$

A_4 :

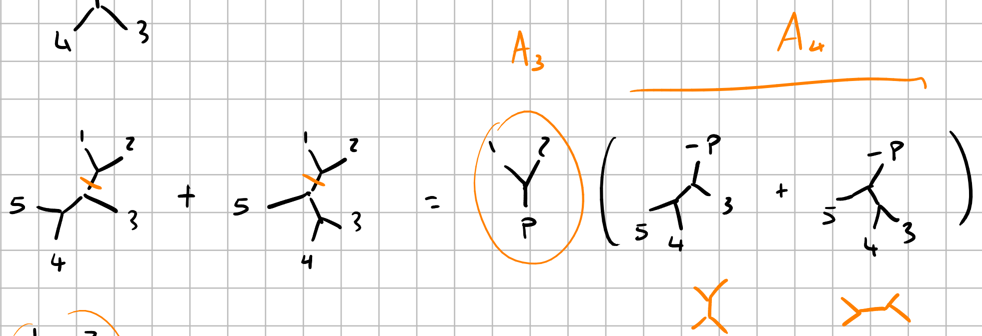


$\text{Res}\, A_4 \;{}_{s_{12}=0} = s_{12}$

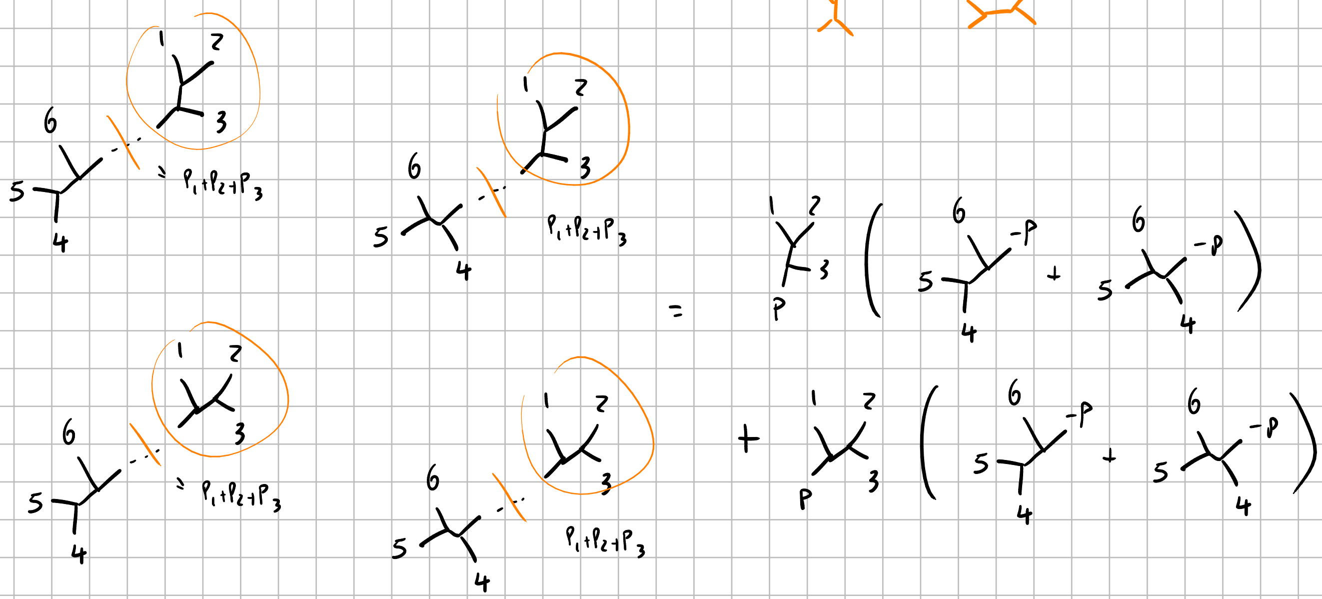


$= 1$

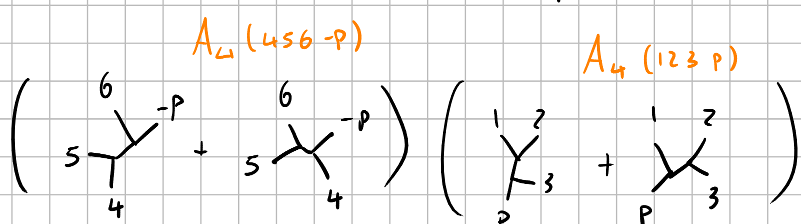
$\text{Res}\, A_{5\,(s_{12}=0)} =$



$\text{Res}\, A_6 (s_{123}=0)$



$=$



BCFW shift:

pick P_i, P_j among the external momenta $(i \neq j)$

send
$$P_i \to P_i + z q \qquad \text{momentum}$$
$$P_j \to P_j - z q$$

q s.t. $q^2 = 0$

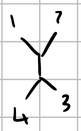
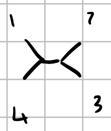
$q \cdot P_i = q \cdot P_j = 0$

ex: 4 particles momenta: $P_1 \; P_2 \; P_3 \; P_4$

after shift: $P_1 + zq \quad P_2 - zq \quad P_3 \quad P_4$

- total momentum is still zero
- $(P_1 + zq)^2 = (P_2 - zq)^2 = 0$

amplitudes now depend on z

$A_4 =$  $+$  $= \frac{1}{s_{12}} + \frac{1}{s_{14}} = \frac{1}{P_1 \cdot P_2} + \frac{1}{P_1 \cdot P_4}$

after shift: $$A_4(z) = \frac{1}{(P_1 + zq)(P_2 - zq)} + \frac{1}{(P_1 + zq)\cdot P_4} = \frac{1}{P_1 \cdot P_2} + \frac{1}{P_1 \cdot P_4 + z P_4 \cdot q}$$

$\Rightarrow = P_1 \cdot P_2$

look at $\frac{A(z)}{z}$ pole at $z = 0$

$$\boxed{A(0) = 2\pi i \oint_{z=0} \frac{A(z)}{z} \, dz}$$

small loop around $z=0$

$$\boxed{X_{1\,j+1}} = s_{12\cdots j} = (P_1 + P_2 + \cdots + P_j)^2 = (P_1 + P_2)^2 + (P_3 + P_4 + \cdots)^2 + 2(P_1 + P_2)\cdot(P_3 + P_4 + \cdots)$$

$$= s_{12} + s_{3\cdots j} + 2P_1 \cdot (P_3 + P_4 \cdots) + 2P_2 \cdot (P_3 + P_4 + \cdots)$$

$$= s_{12} + \underbrace{2 \sum_{i=3}^{j} P_1 \cdot P_i}_{= S_{1j}} + \underbrace{2 \sum_{i=3}^{j} P_2 \cdot P_i}_{2z_j} + s_{3\cdots j}$$

$$= \boxed{s_{12} + \sum_{i=3}^{j} s_{1i} + \sum_{i=3}^{j} s_{2i} + s_{3\cdots j}}$$

$$\boxed{X_{2,j+1}} = (P_2 + \dots + P_j)^2 = (P_3 + P_4 + \dots)^2 + 2P_2 \cdot (P_3 + P_4 + \dots)$$

$$= \boxed{\sum_{i=3}^{j} s_{2i} + s_{3\dots j}}$$

$$\boxed{X_{2n}} = s_{23\dots n-1} = (P_2 + P_3 + \dots P_{n-1})^2 = (P_1 + P_n)^2 = 2P_1 \cdot P_n$$

$$= 2P_1 \cdot (-P_1 - P_2 - \dots - P_{n-1})$$

$$= -2P_1 \cdot P_2 - 2P_1 (P_3 + P_4 + \dots P_{n-1})$$

$$= \boxed{-s_{12} - \sum_{i=3}^{n-1} s_{1i}}$$

(A) set $s_{1i} = 0 \quad (i = 3, \dots, n-1)$

(B) $P_2 \to P_2 + qz$

$P_n \to P_n - qz$

A_5 $\quad X_{13} \quad X_{14} \quad X_{24} \quad X_{25} \quad X_{35}$ $\quad \left(X_{ii} = 0 \quad X_{i\,i+1} = 0 \quad X_{ij} = X_{ji} \right)$

<u>shift:</u> $P_2 \to P_2 + qz \qquad P_5 \to P_5 - qz$

effected if X_{ij} contains either only P_2 or only P_5

X_{13}	effected	$(P_1 + P_2)^2$
X_{14}	effected	$(P_1 + P_2 + P_3)^2 = (P_4 + P_5)^2$
X_{24}	effected	$(P_2 + P_3)^2$
X_{25}	effected	$(P_2 + P_3 + P_4)^2$
X_{35}	uneffected	$(P_3 + P_4)^2 = (P_1 + P_2 + P_5)^2$

---

more generally: if P_2 and P_n shifted:

$X_{1\,j+1}$ always only contains P_2

$X_{2\,j+1}$ always only contains P_2

$X_{3\,j+1}$ contains no P_2 and no P_n

A_5 $\quad X_{13} \quad X_{14} \quad X_{24} \quad X_{25} \quad X_{35}$

zero: $\quad s_{13} = s_{14} = 0$

$$X_{13} = (p_1+p_2)^2 = s_{12} \quad \text{not effected}$$

$$X_{14} = (p_1+p_2+p_3)^2 = \boxed{s_{123}} \xrightarrow{\text{effected}} = \boxed{s_{45}}$$

$$= (\cancel{p_1+p_3})^2 + 2p_2\cdot(p_1+p_3) = \boxed{s_{12}+s_{23}}$$

$\to s_{13} = 0$

$$X_{24} = (p_2+p_3)^2 = s_{23} \quad \text{not effected}$$

$$= (\underbrace{p_4+p_5}+p_1)^2 = (p_1+p_4)^2 + 2p_5(p_1+p_4) = s_{15}+s_{45}$$

$\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad \shortparallel\, 0$

$$X_{35} = (p_3+p_4)^2 = s_{34}$$

$$= (p_1+p_2+p_5)$$

$s_{13} = s_{14} = s_{15} = 0$

$n=6$

$$X_{36} = (p_3+p_4+p_5)^2 = (p_1+p_2+p_6)$$

$$X_{35} = (p_3+p_4)^2 = (p_1+p_2+p_5+p_6)$$

$X^{(0)}_{i\,j+1}$ is the $X_{i\,j+1}$ after setting $s_{1i} = 0$ $(i = 3, \ldots n-1)$

$2 \le j \le n-2$

$$X^{(0)}_{1\,j+1} = s_{12} + \sum_{i=3}^{j} s_{2i} + s_{3\ldots j}$$

$$X^{(0)}_{2\,j+1} = \sum_{i=3}^{j} s_{2i} + s_{3\ldots j}$$

$$X^{(0)}_{2n} = -s_{12}$$

---

after BCFW shift $\quad X_{ij} = X_{ij}(z)$ (polynomial)

$X^{(\infty)}_{ij}$ is the leading order coefficient in z

$$X^{(\infty)}_{ij} = \frac{d}{dz}\Big|_{z=0} X_{ij}(z) \quad \Big\| \quad \text{or} \quad X_{ij}(z) = X_{ij}(0) + X^{\infty}_{ij}\, z + O(z^2)$$

shift P_2 and P_n

$$X_{i\,j+1}(z) = s_{12}(z) + \cancel{\sum_{i=3}^{j} s_{1i}(z)} + \sum_{i=3}^{j} s_{2i}(z) + \cancel{s_{3\ldots j}(z)} = s_{12}(z) + \sum_{i=3}^{j} s_{2i}(z) + O(1)$$

$$s_{12}(z) = 2\, P_1 \cdot (P_2 + q\,z) = 2z\, P_1 \cdot q + O(1)$$

$$s_{2i}(z) = 2\, P_i \cdot (P_2 + q\,z) = 2z\, P_i \cdot q + O(1)$$

$$X_{i\,j+1}(z) = 2z\, P_1 \cdot q + 2z \sum_{i=3}^{j} P_i \cdot q + O(1)$$

$$\begin{cases} X^{(\infty)}_{i\,j+1} = 2\, q \cdot P_1 + 2 \displaystyle\sum_{i=3}^{j} q \cdot P_i \\ X^{(\infty)}_{2\,j+1} = 2 \displaystyle\sum_{i=3}^{j} q \cdot P_i \\ X^{(\infty)}_{2n} = -2\, q \cdot P_1 \end{cases}$$

$j = 2, \ldots, n-2$

$n-2-3+1 = n-3$ eqs

Relations:

$$X^{(\infty)}_{i\,j+1} - X^{(\infty)}_{2\,j+1} + X^{(\infty)}_{2n} = 0$$

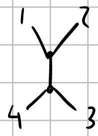
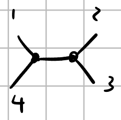
$$X^{(0)}_{i\,j+1} - X^{(0)}_{2\,j+1} + X^{(0)}_{2n} = 0$$

$j = 2, \ldots, n-2$

$(n-3)$ eqs

$$X_{i\,j+1}(z) - X_{2\,j+1}(z) + X_{2n}(z) = O(1)$$

$s_{12} \quad s_{13} \quad s_{14} \quad s_{23} \quad s_{24} \quad s_{34}$

$A_4 = a_1$  $+ \; a_2$ 

impose the hidden zero condition $\Rightarrow$ $a_1 = a_2$

$$\frac{1}{x}(\ldots) + \frac{1}{y}(\ldots)$$

---

goal: show that some of the X_{ij} variables satisfy the same relations if we set a 1-zero or perform a BCFW shift (up to order z)

1-zero: $c_{13} = c_{14} = \cdots = c_{1n-1} = 0 \Rightarrow n-3$ constraints

started with $\frac{n(n-3)}{2}$ variables, now we have $n-3$ less of them

we know that $c_{ij} = X_{ij} + X_{i+1\,j+1} - X_{i+1\,j} - X_{i\,j+1}$

$$0 = X_{13} + X_{24} - \underbrace{X_{23}}_{=0} - X_{14} \qquad \text{since } c_{13} = 0$$

$$0 = X_{14} + X_{25} - X_{24} - X_{15}$$

$$\vdots$$

$$0 = X_{1n-1} + X_{2n} - X_{2n-1} - \underbrace{X_{1n}}_{=0}$$

$$X_{21} = X_{23} = 0$$

$$X_{24} = X_{14} - X_{13}$$

$$X_{25} = X_{24} - X_{14} + X_{15} = X_{15} - X_{13}$$

$$X_{26} = X_{16} - X_{13}$$

$$\vdots$$

$$X_{2n} = X_{1n} - X_{13} = -X_{13}$$

let's do shift instead $\quad P_2 \to P_2 + q z$

$P_n \to P_n - q z$

which X_{ij} are effected by the shift? (assume $i < j$)

$\;\;\;\; \parallel$

$(P_i + P_{i+1} + \dots + P_{j-1})^2$

$i = 1$ or $i = 2$ $\quad X_{1j} \quad j = 3, \dots, n-1$

$X_{2j} \quad j = 4, \dots, n$

are the only effected ones

expand $X_{1j}(z)$ to linear order

$X_{2j}(z)$ to linear order

$$X_{1j}(z) = (P_1 + P_2 + qz + \dots + P_{j-1})^2$$

$$X_{2j}(z) = (P_2 + qz + \dots + P_{j-1})^2$$

$$X_{1j}(z) = (P_1 + P_2 + qz + \dots + P_{j-1})^2$$

$$= (\underline{qz} + \underline{P_1 + P_2 + \dots P_{j-1}})^2$$

$$= (P_1 + P_2 \dots + P_{j-1})^2 + z^2 \cancel{q^2}^{=0} + 2zq \cdot (P_1 + P_2 + \dots + P_{j-1})$$

$$= X_{ij}(0) + 2zq \cdot (P_1 + P_3 + \dots P_{j-1}) \quad \to \text{call it } X_{1j}^{(\infty)}$$

$$= X_{1j}(0) + X_{1j}^{(\infty)} z \qquad z \sim 1 + z$$

$$X_{2j}(z) = (P_2 + qz + \dots + P_{j-1})^2$$

$$= X_{2j}(0) + 2qz \cdot (P_3 + \dots + P_{j-1}) \quad \to X_{2j}^{(\infty)}$$

$$= X_{2j}(0) + X_{2j}^{(\infty)} z$$

$$X_{2j}^{(\infty)} = 2q \cdot (P_3 + \dots + P_{j-1})$$

$$X_{1j}^{(\infty)} = 2q \cdot (P_1 + P_3 + \dots + P_{j-1})$$

$$= \underbrace{2q \cdot P_1}_{= X_{13}^{(\infty)}} + X_{2j}^{(\infty)}$$

$$X_{2j}^{(\infty)} = X_{1j}^{(\infty)} - X_{13}^{(\infty)}$$

$$X_{2j}^{(\infty)} = X_{1j}^{(\infty)} - X_{13}^{(\infty)}$$

$$X_{24}^{(0)} = X_{14}^{(0)} - X_{13}^{(0)}$$

$$X_{25}^{(0)} = X_{15}^{(0)} - X_{13}^{(0)}$$

$$X_{26}^{(0)} = X_{16}^{(0)} - X_{13}^{(0)}$$

$$\vdots$$

$$X_{2n}^{(0)} = X_{1n}^{(0)} - X_{13}^{(0)} = -X_{13}^{(0)}$$

---

let B be an homogeneous rational function of the X_{ij}

a function $f(\lambda x_1, \lambda x_2, \dots, \lambda x_n) = \lambda^n f(x_1, \dots, x_n)$

let B_i be the subset of all terms in B that under the shift $P_2 \to P_2 + zq$ $\quad P_n \to P_n - zq$ scales as z^i as $z \to \infty$

$i \in \mathbb{Z}$

ex:

$$B(z) = \frac{1}{X_{24}(z)} = \frac{1}{X_{2j}(0) + X_{2j}^{(\infty)} z}$$

scales like $\frac{1}{z}$

$\Rightarrow B_{-1}$

$$B(z) = \underbrace{X_{25}(z) + X_{26}(z)}_{B_1(z)} + \underbrace{\frac{X_{35}^2}{X_{24}(z)}}_{B_{-1}(z)} + \underbrace{X_{35}}_{B_0(z)}$$

in general: $\quad B = B_m + B_{m-1} + B_{m-2}$ where z^m is the largest scaling

Prop: these are equivalent:

1. • B satisfies the 1-zero $\overset{def}{\Leftrightarrow}$ (when $X_{2i} = X_{1j} - X_{13}$ holds $B = 0$)

2. • each B_i satisfies the 1-zero

3. • each B_i has enhanced UV scaling z^{i-1}

$z - z$ have UV scaling z^0

---

proof suppose (1) holds

the only way this can happen

→ must be $B = \sum_{c_{ij} \in \text{zero}} c_{ij} A_{ij}$ some function of X_{ij}

$(c_{13}, c_{14}, \ldots c_{1n-1})$

ex: $B = X_{24} - X_{13} + X_{14}$

now let's look at scaling

as $z \to \infty$ $\quad B = z^m B_m(X_{ij}^{(\infty)}) + O(z^{m-1})$

ex: $B(z) = \dfrac{1}{X_{24}(z)} = \dfrac{1}{X_{2j}(0) + X_{2j}^{(\infty)} z}$

$B(z) \to \dfrac{1}{X_{2j}^{(\infty)}} \dfrac{1}{z} = z^{-1} B_{-1}(X^{\infty}_{2j})$

if B has enhanced scaling then

$B_m(X_{ij}^{\infty}) = 0$

↳ the only constraints they satisfy are $X_{2i}^{\infty} = X_{1j}^{\infty} - X_{13}^{\infty}$

B_m satisfies the k-zero

so $B_m = \sum_{c_{ij} \in \text{zero}} c_{ij} \{\ldots\}_{ij}$

$$X_{2j}(0) = X_{1j}(0) - X_{13}(0) \qquad\qquad X_{2j} = X_{2j}(0) + X_{2j}^{\infty} z$$

$$\Updownarrow$$

$$X_{2j}(z) = X_{1j}(z) - X_{13}(z)$$

B satisfies 1-zero then satisfy

as $z \to \infty$ $\quad B(X_{ij}(z)) = 0$

$$= \underbrace{B_m(X_{ij}^{\infty})}_{\to 0} z^m + O(z^{m-1}) \qquad B_m(X_{ij}^{\infty}) = 0$$

B satisfies 1-zero $\Rightarrow$ enhanced scaling of B_m $\Rightarrow$ B_m satisfies 1-zero

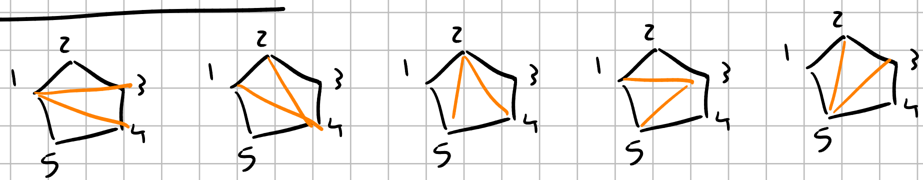
$n=4$ $\qquad B = \frac{1}{X_{13}} + \frac{1}{X_{24}}$ $\qquad$ set $c_{13} = 0$

$$\Updownarrow$$

$$X_{24} = -X_{13}$$

$$B = \frac{1}{X_{13}}$$

$n=5$



suppose

$$B = \underbrace{\underbrace{\frac{a_1}{X_{13}X_{14}} + \frac{a_2}{X_{24}X_{14}} + \frac{a_3}{X_{24}X_{25}}}_{z^{-2}}}_{B_{-2}} + \underbrace{\underbrace{\frac{b_1}{X_{13}X_{35}} + \frac{b_2}{X_{25}X_{35}}}_{z^{-1}}}_{B_{-1}}$$

require that B satisfies the 1-zero

---

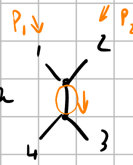
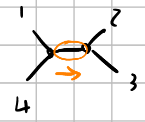
the only term that has $\frac{1}{X_{35}}$ is $\frac{b_1}{X_{13}X_{35}} + \frac{b_2}{X_{25}X_{35}}$

$$0 = B(X^{(0)} \text{ and } X_{35} = 1) = B_{-2} + \frac{b_1}{X_{13}} - \frac{b_2}{X_{13}}$$

$0 = B(X^{(n)} \text{ and } X_{35} = \frac{1}{2}) = B_{-2} + 2\frac{b_1}{x_{13}} - 2\frac{b_2}{x_{13}}$ $\forall X_{ij}$

$\Rightarrow B_{-2}(X^{(n)}) = 0$ and $\frac{b_1}{x_{13}} - \frac{b_2}{x_{13}} = 0$ $\rightarrow b_2 = b_1$

$\Downarrow$

$A_4 = a$  $+ \, b$    generic A_4 within our ansatz

$$= \frac{a}{(p_1+p_2)^2} + \frac{b}{(p_1+p_4)^4} = \frac{a}{X_{13}} + \frac{b}{X_{42}} = \frac{1}{X_{13}}(a-b)$$

impose zero condition: $s_{13} = 0 \iff 0 = X_{13} + X_{24} - \cancel{X_{23}}^{\,0} - \cancel{X_{14}}^{\,0}$

$\downarrow$

if $X_{13} = -X_{24}$ then $A_4 = 0$ (assumption)

recursion relation

$$c_{ij} = X_{ij} + X_{i+1\, j+1} - X_{i+1\, j} - X_{i\, j+1}$$

$$c_{13} = c_{14} = \dots = c_{1\, n-1} = 0 \iff \left.\begin{aligned} X_{24} &= X_{14} - X_{13} \\ X_{25} &= X_{15} - X_{13} \\ &\vdots \\ X_{2n} &= X_{1n} - X_{13} = -X_{13} \end{aligned}\right\} \ n-3 \text{ constraints}$$

Conversely, if we shift
$$p_2 \to p_2 + q z$$
$$p_n \to p_n - q z$$

the only affected X_{ij} are the ones that contain only 1 of p_2 or p_n

X_{1j} $j = 3, \dots, n-1$

X_{2j} $j = 3, \dots, n$

$$X_{1j}(z) = X_{1j}(0) + z X_{1j}^{\infty}$$

$$X_{1j}(z) = (p_1 + p_2 + qz + \dots + p_{j-1})^2 = \cancel{z^2 q^2}^{\,0} + X_{1j}(0) + \underbrace{2 z q \cdot (p_1 + p_2 \dots + p_{j-1})}_{X_{1j}^{\infty}}$$

$$X_{2j}(z) = X_{2j}(0) + \underbrace{2 z q \cdot (p_2 + p_3 + \dots p_{j-1})}_{X_{2j}^{\infty}}$$

$$= X_{2j}(0) + X_{1j}(z) - X_{1j}(0) - 2 z q \cdot p_1 \qquad X_{13}^{\infty}$$

$$X_{13}^{\infty} = 2 q \cdot (p_1 + p_2) = 2 q \cdot p_1$$

$$X^\infty_{2j} = X^\infty_{1j} - X^\infty_{13}$$

Subset zeros $\quad B$ is an homogeneous rational function of the X_{ij}

collect B into subterms depending on how they scale under the shift

$P_2 \to P_2 + zq \qquad P_n \to P_n - zq$

$B(z) = B_m + B_{m-1} + B_{m-2} + \dots$

where B_m contains all terms that scale like z^m (as $z \to \infty$)

Prop. these are equivalent:

- B satisfies 1-zero
- each B_i satisfies 1-zero
- each B_i has enhanced UV scaling z^{i-1}

$\Leftarrow$ obvious

---

step 1 if B satisfies 1-zero then it must have enhanced scaling.

B satisfies zero means: if $X_{2j} = X_{1j} - X_{13}$ then $B = 0$ $\forall$ other choices of X_{ij}

1-zero constraint
$j = 4, \dots n$

$\to$ B must be at least linear in $c_{13}, c_{14}, c_{15}, \dots$

$$B = \sum_{j=3}^{n-1} c_{1j} \{ \dots \}_j$$

---

ex: $B = \dfrac{1}{X_{13}} + \dfrac{1}{X_{24}}$

$= \dfrac{X_{24} + X_{13}}{X_{13}\, X_{24}}$

$0 := -c_{13} = X_{13} + X_{24} - \cancel{X_{14}} - \cancel{X_{23}}$

what happens to B as $z \to \infty$

$$B \to z^m B_m(X_{ij}^\infty) + O(z^{m-1})$$

ex: $B = \frac{1}{X_{13}} + \frac{1}{X_{24}}$ $\qquad X_{24}(z) = X_{24}(0) + z X_{24}^\infty \qquad X_{24}^\infty = -X_{13}^\infty$

$X_{13}(z) = X_{13}(0) + z X_{13}^\infty$

$B = B_{-1}$

$$B(z) = \frac{1}{X_{13}(0) + X_{13}^\infty z} + \frac{1}{X_{24}(0) - X_{13}^\infty z} = \frac{X_{13}(0) + X_{24}(0) + \cancel{X_{13}^\infty z} - \cancel{X_{13}^\infty z}}{-X_{13}^\infty z^2 + - -} \to \frac{1}{z^2}$$

$$= \frac{1}{X_{13}^\infty z} + O(z^{-2}) \qquad \to -\frac{1}{X_{13}^\infty z} + O(z^{-2})$$

B has enhanced scaling

$$\frac{1}{X_{13}}\Big|_{X_{13} = X_{13}^\infty} + O(z^{-2})$$

$f(z) = z$ $\qquad f + g = 0$

$g(z) = -z$

$$B = \frac{1}{1+z} + \frac{1}{2-z} = \frac{2 - z + 1 + z}{(1+z)(2-z)} = \frac{3}{- - - z^2}$$

Laurent series at ∞?

$f(z) = \frac{1}{a + bz}$ expand at ∞

change of variable $z = 1/w$

expand at $w = 0$

$$f = \frac{1}{a + b/w} = \frac{w}{wa + b} = \frac{w}{b} + O(w^2) \quad \text{(taylor series)}$$

$$f'(0) = \frac{(wa+b) - wb}{(wa+b)^2}\Big|_{w=0} = \frac{b}{b^2} = \frac{1}{b}$$

$$\Rightarrow \frac{1}{bz} + O(z^{-2})$$

---

as $z \to \infty$ $\quad B_m(z) \to z^m B_m(X_{ij}^\infty) + O(z^{m-1})$ $\qquad$ Taylor series around $z = \infty$

B_m has enhanced scaling $\iff B_m(X_{ij}^\infty) = 0 \quad \forall X_{ij}^\infty$

X_{ij}^∞ satisfies the same relations as 1-zero

$B_m(X_{ij}^\infty) = 0 \quad \forall X_{ij}^\infty \iff B_m$ satisfies 1-zero

this proves that ② ⟺ ③ and ② ⟹ ①

suppose ① is true

want to show

B satisfies 1-zero $\Rightarrow$ B_m satisfies 1-zero

if we let $X_{2j}(0) = X_{1j}(0) - X_{13}(0)$ then $B = 0$

$X_{ij}(0) + z X_{ij}^\infty$

then we do shift then $X_{2j}(z) = X_{ij}(z) - X_{13}(z)$

$X_{2j}(0) + z X_{2j}^\infty$

$B(X_{ij}(z))$

then $B(z) \to 0$ as $z \to \infty$

since $B = z^n B_m(X_{ij}^\infty) + O(z^{m-1})$ it must be $B_m(X_{ij}^\infty) = 0$

---

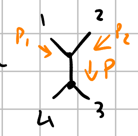
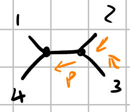
let $B' = B - B_m = B_{m-1} + \dots$

ex: $B(x,y) = x - y$

$x = 1 + z$
$y = -1 - z$

$B(1+z, -1-z) = 0$

4 point

$B =$ a  $+\ b$ 

$$= \frac{a}{X_{13}} + \frac{b}{X_{24}}$$

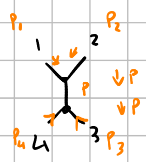
$X_{13}\,,\ X_{24}$

independent planar variables

$$X_{ij} = (p_i + p_{i+1} + \dots + p_{j-1})^2$$

$$X_{ii} = X_{i\,i+1} = 0$$

$$X_{in} = 0$$



each vertex $\propto \delta$ of incoming − outgoing momentum

each internal line $= \int dp \, \frac{1}{p^2}$ of associated momentum

$$\int dp \; \underline{\delta(p_1+p_2-p)} \; \overbrace{\delta(p_3+p_4+p)\,\frac{1}{p^2}}^{f(p)} = \frac{\delta(p_1+p_2+p_3+p_4)}{(p_1+p_2)^2} = \frac{\delta(p_1+p_2+p_3+p_4)}{X_{13}}$$

$$\int dp \; \delta(p-q)\, f(p) = f(q)$$

---

then impose 1-zero $\qquad c_{13} = 0 \qquad (n=4) \qquad c_{13} = c_{14} = \dots = c_{1\,n-1} = 0$

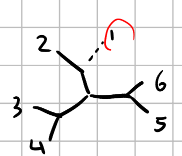
require that $c_{13}=0 \Rightarrow B = 0 \ \forall X$ $\qquad$ $(n-3)$ constraints

$$c_{13} = 0$$
$$\parallel$$
$$X_{13} + X_{24} - \cancel{X_{14}}^{\,0} - \cancel{X_{34}}^{\,0}$$

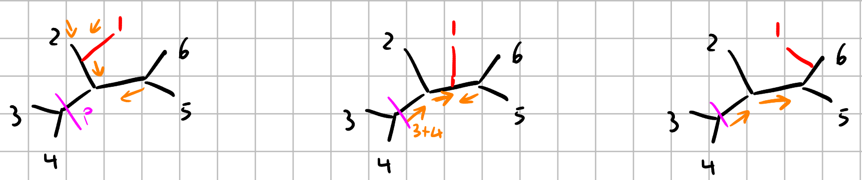
$$B = \frac{a}{X_{13}} + \frac{b}{X_{24}} = \frac{a-b}{X_{13}} \qquad \text{so} \quad \underline{a = b}$$

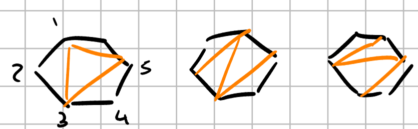
---

ex: $n = 6$ $\qquad$ choose a 6-leg diagram with leg 1 removed

start with 

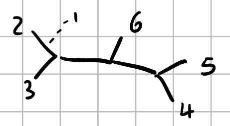
add leg 1 back in in all possible ways (according to order)

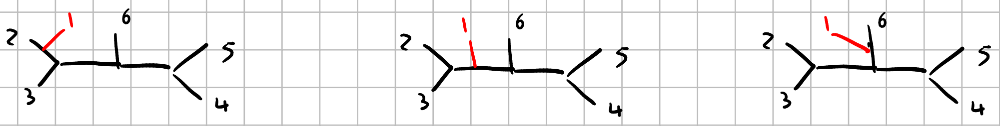




$$S_1 = \frac{a_1}{X_{13}\,X_{51}\,X_{35}} + \frac{a_2}{X_{25}\,X_{15}\,X_{35}} + \frac{a_3}{X_{25}\,X_{26}\,X_{35}}$$

another D-subset

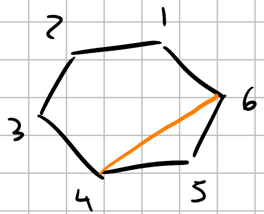


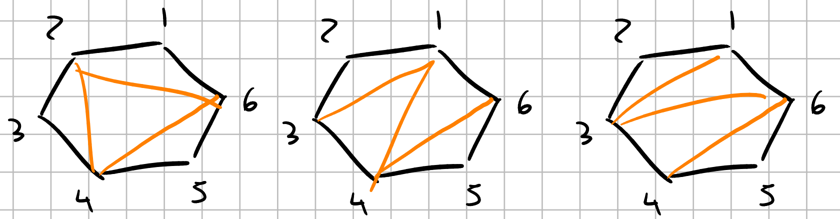


$$S_2 = \frac{a_4}{X_{13}\,X_{46}\,X_{41}} + \frac{a_5}{X_{24}\,X_{46}\,X_{41}} + \frac{a_6}{X_{24}\,X_{46}\,X_{26}}$$

$S_1 + S_2$ are the diagrams that contain X_{46} and X_{35}

<u>all diagrams</u> that have $X_{46} = (p_4 + p_5)^2$





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look at $S_1 + S_2$ and ask that it satisfies 1-zero ($c_{13} = c_{14} = c_{15} = 0$)

$\gg$ $X_{2j} = X_{1j} - X_{13}$ $\quad j = 4, \dots, 6$

if $c_{13} = c_{14} = c_{15} = 0 \Rightarrow S_1 + S_2 = 0 \quad \forall X_{ij}$ subject to constraint

<u>then</u> $c_{13} = c_{14} = c_{15} = 0 \Rightarrow S_1 = 0$ and $S_2 = 0 \quad \forall X_{ij}$

$$S_1 + S_2 = \frac{a_1}{X_{13}\,X_{51}\,X_{35}} + \frac{a_2}{X_{25}\,X_{15}\,X_{35}} + \frac{a_3}{X_{25}\,X_{26}\,X_{35}}$$

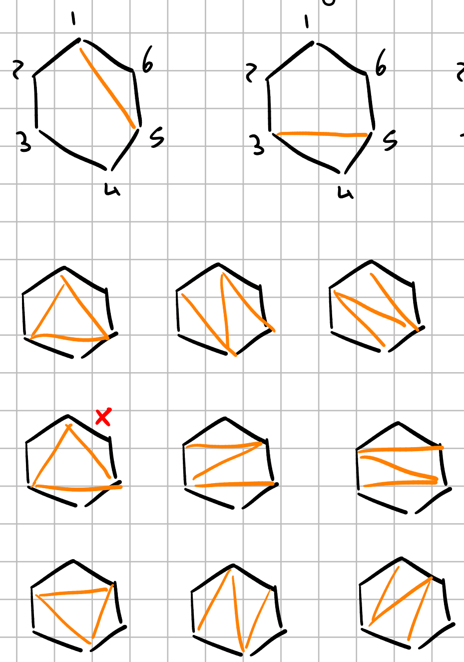
$$+ \frac{a_4}{X_{13}\,\boxed{X_{46}}\,X_{41}} + \frac{a_5}{X_{24}\,\boxed{X_{46}}\,X_{41}} + \frac{a_6}{X_{24}\,\boxed{X_{46}}\,X_{26}}$$

$$= \frac{1}{X_{35}}\underbrace{(\,--\,)}_{=0} + \frac{1}{X_{46}}\underbrace{(\,--\,)}_{=0}$$

$X_{2j} = X_{1j} - X_{13}$

all possible diagrams with 6 legs

group them by



splitting depending on z scaling of propagators

boundary propagators: the ones that scale linearly with z under BCFW shift (X_{1j} and X_{2j})

group by # of boundary propagators

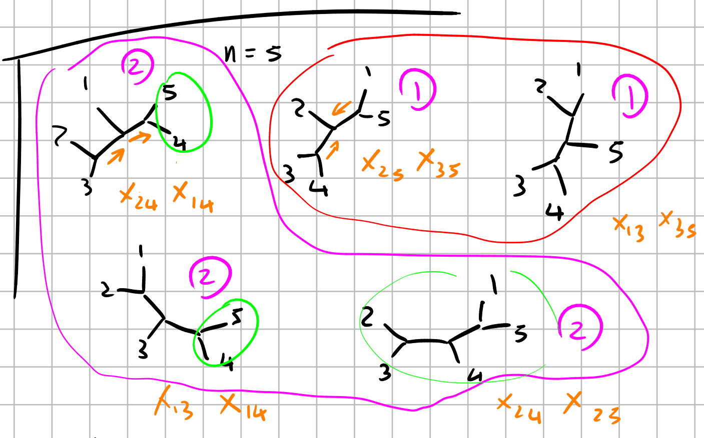
→ group them by scaling under BCFW shift

$$B = \boxed{B_0} + \boxed{B_{-1}} + \boxed{B_{-2}} + \dots$$

B_0 ↓ 0 boundary propagators

B_{-1} ↓ 1 bound. prop.

B_{-2} ↓ 2 boundary prop.



bound. prop.: X_{13} X_{14} X_{24} X_{25}

1-zero and BCFW shift

up to cyclic permutations a 1-zero is the imposition of the $n-3$ constraints

$c_{13} = c_{14} = \dots = c_{1,n-1} = 0$ on the kinematic data.

Since $c_{ij} = X_{ij} + X_{i+1,j+1} - X_{i\,j+1} - X_{i+1,j}$

the constraints can be equivalently be written as

$$\begin{cases} X_{13} + X_{24} - X_{14} - \underset{=0}{X_{23}} = 0 \\ X_{14} + X_{25} - X_{15} - X_{24} = 0 \\ \vdots \\ X_{1,n-1} + X_{2n} - \underset{=0}{X_{1n}} - X_{2\,n-1} = 0 \end{cases} \iff \begin{cases} X_{24} = X_{14} - X_{13} \\ X_{25} = X_{15} - X_{13} \\ \vdots \\ X_{2n} = X_{1n} - X_{13} = -X_{13} \end{cases}$$

so the 1-zero fully fixes X_{2j} once X_{1j} is specified, and does not affect X_{ij} if $i,j \notin \{1,2\}$.

---

the corresponding BCFW shift is the change $\begin{cases} P_2 \to P_2 + zq \\ P_n \to P_n - zq \end{cases}$

where z is a complex parameter and q satisfies $\begin{cases} q^2 = 0 \\ q \cdot P_2 = q \cdot P_n = 0 \end{cases}$

so that
$(P_2 + zq)^2 = 0$
$(P_n - zq)^2 = 0$

Let $X_{ij}(z)$ be the z-dependent planar invariant.

$X_{ij}(z)$ depends on z only if it includes exactly <u>one</u> of P_2, P_n in the sum, otherwise either there is no z or they cancel out.

$\to$ the only z-dependent ones are $X_{1j}(z)$ $j = 3, \dots, n-1$

and $X_{2j}(z)$ $j = 4, \dots, n$.

Let X^∞_{ij} denote the coefficient of the largest power of z appearing in $X_{ij}(z)$.

$$X_{1j}(z) = (P_1 + P_2 + zq + \dots + P_{j-1})^2 = (P_1 + P_2 + \dots + P_{j-1})^2 + \underbrace{(zq)^2}_{q^2=0} + 2\,zq\cdot(P_1 + P_2 + \dots + P_{j-1})$$

$$= X_{1j}(0) + z\;2q\cdot(P_1 + P_2 + \dots + P_{j-1}) \quad \Rightarrow \quad X_{1j}^{\infty} = 2q\cdot(P_1 + P_2 + \dots P_{j-1}) \quad (j = 3, \dots, n-1)$$

$$X_{2j}(z) = (P_2 + zq + \dots + P_{j-1})^2 = \dots$$

$$= X_{2j}(0) + z\;2q\cdot(P_2 + P_3 + \dots + P_{j-1}) \quad \Rightarrow \quad X_{2j}^{\infty} = 2q\cdot(P_2 + P_3 + \dots + P_{j-1}) \quad (j = 4, \dots, n)$$

in all other cases $X_{ij}(z) = X_{ij}(0)$ so $X_{ij}^{\infty} = X_{ij}(0)$

note that $X_{2j}^{\infty} = X_{1j}^{\infty} - 2q\cdot P_1 = X_{1j}^{\infty} - X_{13}^{\infty}$ $\quad (j = 4, \dots, n)$

so the X_{ij}^{∞} satisfy the same constraint as the 1-zero.

Subset zeros

if $\lambda \neq 0$ $\quad B(\lambda X_{ij}) = \lambda^s B(X_{ij})$ for some $s \in \mathbb{Z}$

Let $B(X_{ij})$ be an homogeneous rational function of the X_{ij} variables.

We will write $B(X) = B_m(X) + B_{m-1}(X) + \dots$

where $B_i(X)$ is the sum of all the terms of B that scales like z^m as $z \to \infty$ under the BCFW shift. (There are finitely many B_i)

example: $n = 5$, independent X_{ij} are $X_{13}, X_{14}, X_{24}, X_{25}, X_{35}$

scale like z $\qquad$ scales like z^0

if $B = \dfrac{X_{13}}{X_{24}X_{14}} + \dfrac{X_{35}}{X_{13}X_{24}} + \dfrac{1}{X_{35}}$ (note it's homogeneous)

scales like $\dfrac{z}{z^2} = \dfrac{1}{z}$ $\qquad$ scales like $\dfrac{1}{z^2}$ $\qquad$ scales like z^0

then $B_0 = \dfrac{1}{X_{35}}$ $\quad B_{-1} = \dfrac{X_{13}}{X_{24}X_{14}}$ $\quad B_{-2} = \dfrac{X_{35}}{X_{13}X_{24}}$ and $B = B_0 + B_{-1} + B_{-2}$

Def: we say a function $B = B(X_{ij})$ satisfies the 1-zero if

$X_{2j} = X_{1j} - X_{13} \Rightarrow B(X) = 0$ ($\forall$ choices of X_{ij} with $i,j \neq 2$)

45 mins

10:25

Prop. The following are equivalent.

① B satisfies the 1-zero

② each B_i satisfies the 1-zero

Proof ② $\Rightarrow$ ① is trivial.

Suppose ① is true. and let $B(z) = B(X_{ij}(z))$

note that as $z \to \infty$ $B(z) = z^m B_m(X_{ij}^\infty) + O(z^{m-1})$

This is because B is a rational function and m is the largest power of z appearing.

ex: if $B = \underbrace{\dfrac{X_{13}}{X_{24}X_{14}}}_{B_{-1}} + \underbrace{\dfrac{X_{35}}{X_{13}X_{24}}}_{B_{-2}}$

then $B(z) = B_{-1}(z) + O(z^{-2})$

$$= \frac{X_{13}(0) + z X_{13}^\infty}{(X_{24}(0) + z X_{24}^\infty)(X_{14}(0) + z X_{14}^\infty)} + O(z^{-2})$$

$$= \frac{1}{z} \frac{X_{13}^\infty}{X_{24}^\infty X_{14}^\infty} + O(z^{-2}) = z^{-1} B_{-1}(X_{ij}^\infty) + O(z^{-2})$$

since $X_{2j}^\infty = X_{1j}^\infty - X_{13}^\infty$

note that if $X_{2j}(0) = X_{1j}(0) - X_{13}(0)$ then $X_{2j}(z) = X_{1j}(z) - X_{13}(z)$ $\forall z$

So $B(z)$ satisfies the zero $\forall z$. But then

$0 = z^m B_m(X_{ij}^\infty) + O(z^{m-1})$ $\forall z$ so it must be $B_m(X_{ij}^\infty) = 0$

the only constraints on the X_{ij}^∞ are the same as the 1-zero,

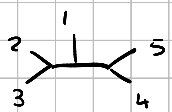
so this means B_m satisfies the 1-zero.

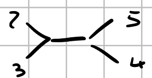
Then we proceed by induction. if B satisfies the 1-zero and B_m satisfies the 1-zero then $B' = B - B_m = B_{m-1} + B_{m-2} + \dots$ satisfies the 1-zero. Following the same steps as above we then prove that B_{m-1} satisfies the 1-zero and so on.

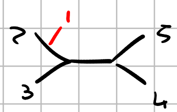
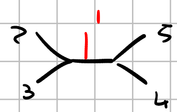
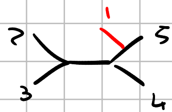
Generic amplitude: arbitrary linear combination of all ordered Feynman diagrams for n external legs.

D-subsets

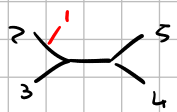
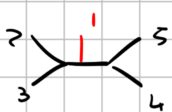
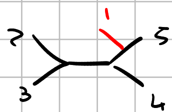
A D-subset at n-point is the subset of the generic amplitude that contains all diagrams obtained by starting with a particular Feynman diagram, removing leg 1, then reattaching it in all possible ways (within the order).

For example: $n=5$, start with 

- remove leg 1: $\rightarrow$ 

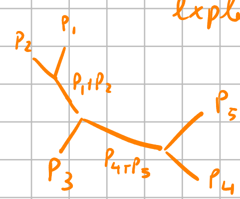
- add back leg 1 in all ways:   

the associated D-subset is

a_1  $+\ a_2$  $+\ a_3$ 

$$= \frac{a_1}{X_{13}\,X_{41}} + \frac{a_2}{X_{24}\,X_{41}} + \frac{a_3}{X_{24}\,X_{52}}$$

explanation

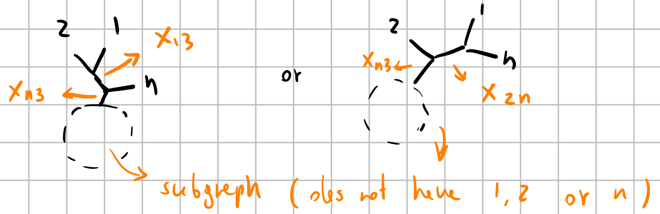


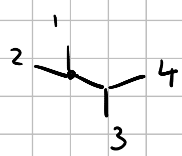
$$= \frac{1}{(p_1+p_2)^2}\,\frac{1}{(p_4+p_5)^2} = \frac{1}{X_{13}}\,\frac{1}{X_{41}}$$

Def: A "boundary propagator" is an X_{ij} that depends on z under a BCFW shift (so $X_{13}, \dots, X_{1n-1}$ and $X_{24}, \dots, X_{2n}$)

Now, if B is our generic amplitude at n points, we split it in groups depending on the number of boundary propagators each term has. This is the same as splitting $B = B_m + B_{m-1} + \dots$ from earlier, since a term with k boundary propagators scales as z^{-k} when $z \to \infty$.

• A graph with 1 boundary propagator looks like



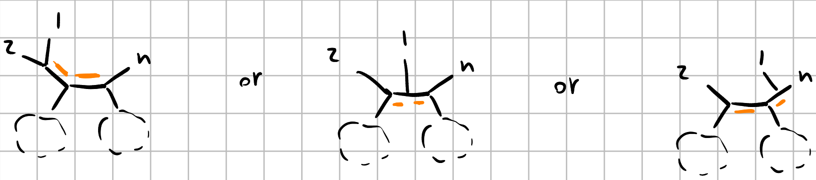


$= \frac{1}{x_{12}}$ $n = 4$ (only 1)

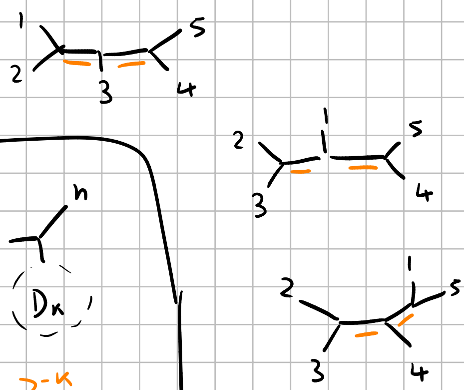
$n = 5$

$\frac{1}{x_{13}\,x_{35}}$ $\frac{1}{x_{25}\,x_{35}}$

• 2 boundary operators

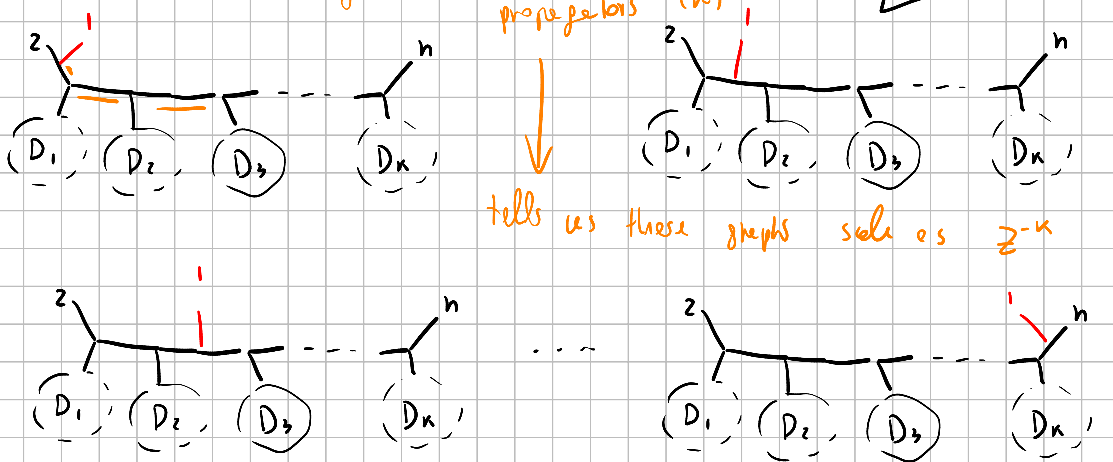


$n = 5$ 2 boundary



• k boundary operators:

orange lines count # boundary propagators (k)



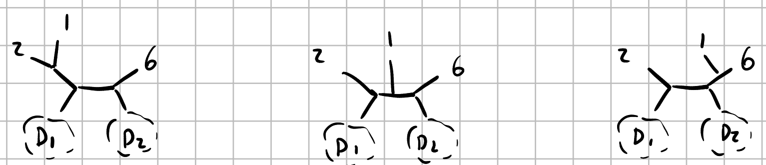
tells us these graphs scale as z^{-k}

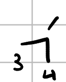
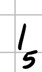
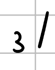
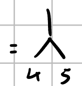
It should be clear from the image that removing leg 1 and re-adding it does not change the # of boundary propagators, so we can partition each B_{-k} in D-subsets (by going through all choices of subgraphs $D_1, \dots D_k$).

Let's see an example:

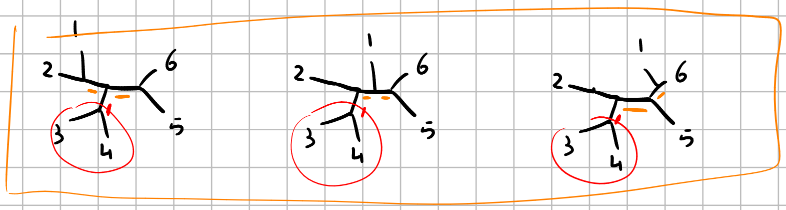
$n = 6$, 2 boundary propagators.

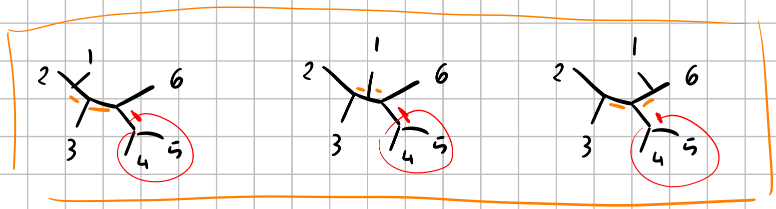
must look like



the choices for D_1, D_2 are: ($D_1 =$  $D_2 =$ ) and ($D_1 =$  $D_2 =$ )

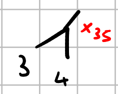
So there are 2 D-subsets:

$S_1 =$ 

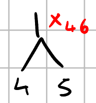
$S_2 =$ 

Since $S_1 + S_2 = B_{-2}$ we know it must satisfy 1-zero if B does.

However, so do S_1 and S_2 independently, since

S_1 has a factor of X_{35} coming from the 

piece

and S_2 has a factor of X_{46} coming from the 

piece.

$$\rightarrow \quad S_1 + S_2 = \frac{1}{X_{35}}(--) + \frac{1}{X_{45}}(---) = 0 \quad \forall\, X_{35}, X_{45} \text{ only if}$$

the coefficients of $\frac{1}{X_{35}}$ and $\frac{1}{X_{45}}$ vanish independently $\Rightarrow S_1 = 0,\ S_2 = 0$.

Moreover, imposing the 1-zero uniquely determines S_1 and S_2 up to an overall coefficient. Note that

$$S_1 = \frac{1}{X_{35}}\left(\frac{a_1}{X_{13}\,X_{15}} + \frac{a_2}{X_{25}\,X_{15}} + \frac{a_3}{X_{25}\,X_{26}} \right) = 0$$

$$\Leftrightarrow \quad \frac{a_1}{X_{13}\,X_{15}} + \frac{a_2}{X_{25}\,X_{15}} + \frac{a_3}{X_{25}\,X_{26}} = 0$$

using the fact that the 1-zero constrains $X_{26} = -X_{13}$

we get
$$0 = \frac{a_1}{X_{13}\,X_{15}} + \frac{a_2}{X_{25}\,X_{15}} - \frac{a_3}{X_{25}\,X_{13}}$$

$$= \frac{e_1 X_{25} + e_2 X_{13} - e_3 X_{15}}{X_{13} X_{15} X_{25}} = 0$$

$$X_{25} = X_{15} - X_{13}$$

so it must be $e_1 X_{25} + e_2 X_{13} - e_3 X_{15} = 0$

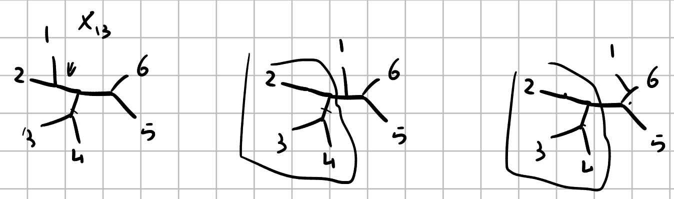
$$= e_1 (X_{15} - X_{13}) + e_2 X_{13} - e_3 X_{15} = 0$$

$$= X_{15} (e_1 - e_3) + X_{13} (e_2 - e_1)$$

Since X_{13}, X_{15} are arbitrary it must be $e_1 = e_2$ and $e_1 = e_3$

so $e_1 = e_2 = e_3$.

alternative



x_{25}

X_{2j} are arbitrary and X_{1j} fixed by X_{2j}

$$\left(\begin{array}{l} X_{24} = - X_{13} \\ X_{2j} = X_{1j} - X_{13} \end{array} \right.$$

$$X_{1j} = X_{2j} + X_{26}$$

since X_{25} arbitrary

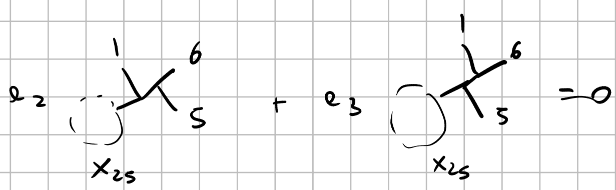
$$S_1 = (- -)_1 + \frac{1}{X_{25}} (-)_2 = 0 \Rightarrow \begin{array}{l} (- -)_1 = 0 \\ (- -)_2 = 0 \end{array}$$

eq 22 of Rodina

idea:

$$\left\{ \begin{array}{l} a + b = 0 \\ a + 2b = 0 \end{array} \right. \Leftrightarrow \{$$

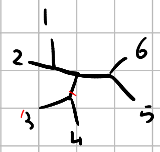
using that

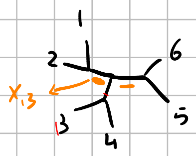


$$\frac{1}{X_{25}} \left(\frac{e_2}{X_{15}} + \frac{e_3}{X_{26}} \right) = 0$$

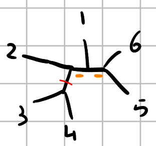
$$X_{26} = X_{16} - X_{13}$$

Looks like eq. 22 is wrong (which is good!)

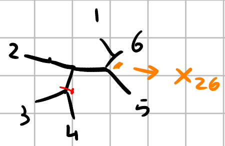
(otherwise the coefficient of  would be zero)



$$\frac{1}{X_{13}\,X_{15}}$$

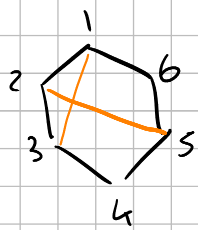


$$\frac{1}{X_{25}\,X_{15}}$$



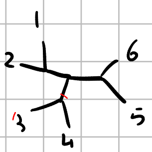
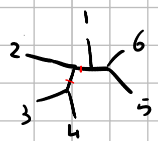
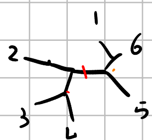
$$\frac{1}{X_{25}\,X_{26}} \rightarrow -\frac{1}{X_{25}\,X_{13}}$$

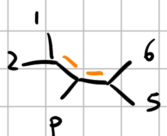
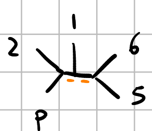
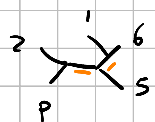
$X_{26} = -X_{13}$



intersecting chords, not a triangulation

this does not correspond to a Feynman diagram

$S_1 = e_1$  $+\; e_2$  $+\; e_3$ 

$= \dfrac{1}{X_{35}} \Big(e_1$  $+\; e_2$  $+\; e_3$  $\Big)$

all graphs with $n=5$ that have 2 boundary propagator